

Bayesian Structural Inference for Dynamic Crowdsourcing Contests

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Online crowdsourcing contests generate rich dynamic behavior, but their core strategic state is latent: contestants react to noisy public leaderboard feedback rather than true relative performance. We develop a tractable dynamic game with Bayesian filtering and a corresponding structural estimation framework to recover latent competition state, effort dynamics, participant ability, innovation volatility, and signal precision. The model admits a closed-form Markov Perfect Equilibrium, enabling disciplined Bayesian estimation in a high-frequency dynamic setting. For validation, we use a holdout design: private-leaderboard path values are excluded from the estimation likelihood and reserved for diagnostic tests, although final private ranks are used only to define the focal rivalry. Contestants do not observe this leaderboard during the contest, but it is generated from the same submissions and scoring rule as the public leaderboard. Using data from 73 Kaggle contests, we find that contests with higher estimated innovation risk have more volatile private-leaderboard paths. For signal precision, the evidence is weaker but directionally consistent: higher estimated precision is associated with smaller public-private leaderboard disagreement. These results provide disciplined diagnostic evidence on the estimated primitives and illustrate how the estimated model can be used to study counterfactual changes in prize levels and contest duration.

Key words: Bayesian Statistics, Structural Estimation, Crowdsourcing Contest, Dynamic Contest

1. Introduction

Crowdsourcing contests are widely used by firms to source solutions to complex problems from external contributors. Platforms such as Kaggle¹, Topcoder², DrivenData³, and AICrowd⁴ host thousands of these contests, attracting data scientists, engineers, and developers to compete by submitting solutions to real-world problems. On Kaggle, for instance, a typical contest spans several weeks, during which participants compete for monetary prizes by repeatedly submitting solutions, observing their relative ranking on a public leaderboard, and adjusting their strategies in response to evolving feedback and the observed actions of competitors.

The central empirical challenge in these online crowdsourcing contests is that the true competitive state is unobserved during the contest. Participants observe noisy public leaderboard updates rather than true relative performance, and they choose effort based on filtered beliefs rather than on the latent state itself. The econometrician faces an analogous partial-observability problem during estimation, making it difficult to recover strategic behavior and incentives from contest-phase data alone.

Motivated by this identification problem, we develop a dynamic contest model and structural Bayesian estimation framework tailored to settings such as Kaggle, where contestants choose continuous effort while learning from noisy public leaderboard signals. Applying the framework to the Meta-Kaggle dataset (Risdal and Bozsolik 2022), we recover latent competition states and structural primitives from contest-phase public information and evaluate them using post-contest private-leaderboard paths excluded from the likelihood.

To illustrate the framework in greater detail, Figure 1 depicts the key components and their interactions in a typical Kaggle contest. One notable feature is the existence of two leaderboards: a *private* leaderboard, which is hidden until the contest ends and ultimately determines the final ranking and prize allocation, and a *public* leaderboard, which is visible to all participants during the contest.⁵ Leaderboard displays are contestant-specific and update asynchronously. After observing the public leaderboard, contestants update their beliefs and form a perceived state of the competition. Based on this belief, they choose their effort level, which corresponds to a *Markov Perfect Equilibrium* (MPE). These effort levels shape submission intensity. Each submission generates a new score for the submitting contestant, while the displayed score of the opponent remains

¹ Kaggle: <https://www.kaggle.com>

² Topcoder: <https://www.topcoder.com>

³ DrivenData: <https://www.drivendata.org>

⁴ AICrowd: <https://www.aicrowd.com>

⁵ This design serves two purposes. First, the public leaderboard provides participants with real-time feedback. Second, by keeping the private leaderboard concealed, the platform discourages overfitting to the public metric and preserves uncertainty, thereby sustaining competitive tension throughout the contest.

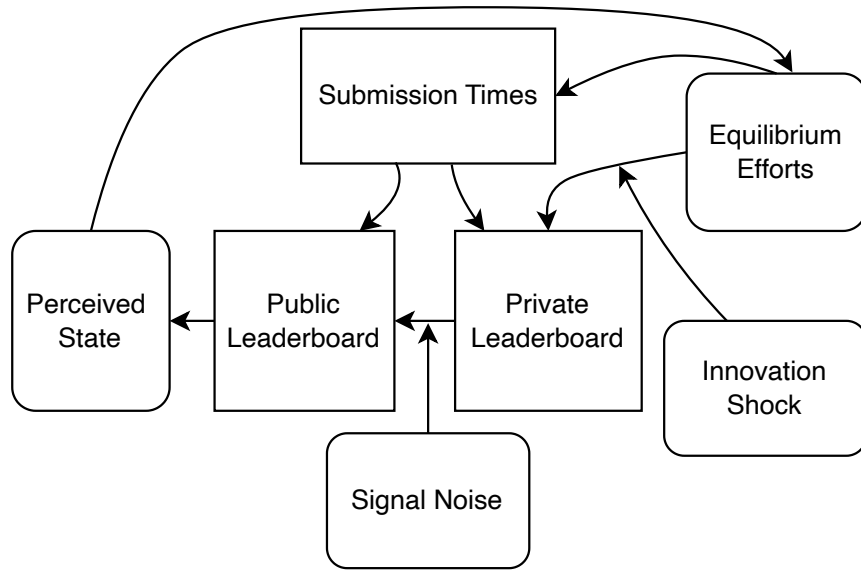


Figure 1 Structure of a Typical Kaggle Contest

Note: This figure illustrates the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles and rounded rectangles represent observed data and latent states, respectively. The private leaderboard is visible only to the contest organizer during the contest and is revealed to all participants only after it ends. During the contest period, contestants can observe only the submission times of each player and the public leaderboard.

at its most recent submitted value. The private leaderboard, in particular, evolves according to the real-time effort level and an innovation shock.

We estimate the structural parameters of the model using a Bayesian approach, based on contestants' submission times and the observed *public* leaderboard trajectories from the Meta-Kaggle database. The estimation relies solely on information available to participants during the contest and leverages the equilibrium structure implied by the dynamic model. This procedure recovers latent competition states, effort dynamics, participant cost parameters, innovation uncertainty, and the precision of the public measurement process. Under the maintained steady-state approximation, signal precision primarily governs filtering and the observation system rather than equilibrium-effort comparative statics.

To assess whether the estimated model captures meaningful contest primitives, we use private-leaderboard paths to evaluate two predictions illustrated in Figure 1: higher estimated innovation risk should be associated with greater private-path volatility, whereas higher estimated signal precision should be associated with smaller public-private leaderboard gaps. Although private-leaderboard path values are excluded from estimation, final private ranks are used to select the focal pair of contestants. We therefore interpret the results as diagnostic evidence conditional on the selected pair, rather than as validation based on a fully independent sample. To strengthen the

credibility of these diagnostics, we also conduct time-split exercises that relate primitives estimated from early public contest data to private-path outcomes in the later contest period, thereby reducing the concern that the full-path relationships merely reflect common path-level variation.

Empirically, we carefully select 73 high-quality contests from the Meta-Kaggle database and preprocess the data to ensure consistency across estimation methods. Conditional on the focal-pair selection rule, we estimate contest-specific innovation risk and signal precision using only public contest-phase information. The validation results show that structurally estimated innovation risk is positively associated with private-leaderboard volatility, including in a time-split diagnostic. For signal precision, the evidence is weaker but directionally consistent: higher estimated signal precision is associated with smaller public-private leaderboard disagreement, and the positive relationship between structural and benchmark precision estimates also appears under an alternative focal-pair specification. Accordingly, our empirical interpretation of λ is primarily as a measurement and filtering primitive.

This paper makes two main contributions. First, we develop a dynamic game-theoretic model and a Bayesian structural framework for contests with noisy public feedback and latent performance dynamics. Relative to contest models centered on discrete submit-or-stop decisions, our approach models continuous effort accumulation and makes belief updating from noisy signals structurally central. Building on the equilibrium structure in [Ryvkin \(2022\)](#), we add noisy public signals and Bayesian filtering while retaining a closed-form equilibrium for estimation. Second, we use private-leaderboard path values, which are excluded from the likelihood, to construct post-estimation validation targets. Because final private ranks enter focal-pair selection, we interpret these exercises as path-wise holdout diagnostics conditional on ex post pair selection. This provides a disciplined assessment of whether the estimated structure captures innovation uncertainty and signal precision rather than merely fitting observed contest-phase behavior. The results provide post-estimation diagnostic evidence on the recovered contest primitives and highlight the value of the model for mechanism evaluation and platform design. Taken together, these contributions advance our understanding of online competitive dynamics and provide a foundation for future empirical work on digital contest design.

1.1. Related Literature

This study is most closely related to the following strands of literature.

Contest Design. A rich literature on *static* contest design ([Ales et al. 2017](#)) investigates how to structure competitive environments to elicit performance, typically through a contest success function mapping effort to winning probabilities ([Skaperdas 1996](#)).⁶ Because our setting is dynamic

⁶ There are exceptions, such as beauty or barrier contests.

with repeated feedback, this static branch serves mainly as background for the mechanism-design questions we build on, including prize allocation, participation rules, and information disclosure (Moldovanu and Sela 2001, Ales et al. 2021, Bergemann et al. 2022).

While these studies offer valuable insights into incentive design, they largely overlook how participants dynamically adjust their efforts in response to evolving competition. In contrast, this paper focuses on *dynamic* contests, where strategic decisions unfold over time. Two main modeling approaches have been used to capture such dynamic strategic interactions:

One strand of the literature models dynamic contests as multistage processes, where participants exert effort over time while facing uncertainty about intermediate progress. This structure facilitates the analysis of the mid-contest policy instruments available to the contest designer, such as feedback disclosure and screening mechanisms. Specifically, in a typical two-stage setting (Aoyagi 2010, Mihm and Schlapp 2019), the initial stage captures either feasibility uncertainty or foundational experimentation, with progress revealed through milestone feedback provided by the contest organizer. Participants exert effort in both stages, and the final output reflects the cumulative impact of early performance and subsequent improvement. Similarly, Bimpikis et al. (2019) study a two-stage model in which agents exert continuous effort to complete Stage A (potentially infeasible) and Stage B (feasible), with breakthroughs governed by Poisson processes. In all these models, the contest designer may strategically disclose interim feedback. Another example is Khorasani et al. (2023), who analyze a two-stage contest in which solvers invest exploratory effort to generate viable ideas, with only those passing an imperfect screening mechanism proceeding to execution.

The second approach, more closely related to our work, treats output as a stochastic process whose drift depends on the participant’s effort. This line of research originates from the tug-of-war framework, first formalized by Harris and Vickers (1987) as a one-dimensional simplification of a multi-stage R&D race. Subsequent studies build on this foundation by modeling each player’s progress as a Brownian motion. Budd et al. (1993), for example, analyze a duopoly innovation race where the state evolves as a Brownian motion driven by the difference in effort, and characterize an approximate equilibrium. Moscarini and Smith (2011) advance this approach by modeling the game state as the output gap between players and derive closed-form equilibrium strategies under pure-strategy play. Ryvkin (2022) adapt their framework to a dynamic contest with a fixed deadline, providing a closed-form solution for the Markov perfect equilibrium.

This paper studies online crowdsourcing contests, which typically feature fixed deadlines, repeated submissions, and public real-time feedback, all governed by platform-specific rules and evaluation metrics. Hence, our analysis builds on the model by Ryvkin (2022), whose simplifications yield a closed-form equilibrium that makes structural estimation in a dynamic setting analytically and computationally tractable. We depart from their full-information setting by introducing a noisy

public signal and Bayesian filtering, thereby bringing partial observability into a tractable dynamic contest model.

Structural Estimation of Contests. Beyond theoretical modeling, a growing body of research employs structural estimation techniques to recover key parameters of contest environments using observed data.

In the structural estimation of static contests, when a contest success function (CSF) is explicitly specified, the primary goal is to estimate its functional form or key parameters (Hwang 2012, Kang 2016, Huang and He 2021). In contrast, for contests without a well-defined or smooth CSF, such as beauty contests, researchers typically rely on equilibrium conditions, most notably the first-order condition. For example, Yoganarasimhan (2016) proposes a two-step semiparametric estimator, resembling the EM algorithm, to recover both unobserved auction heterogeneity and bidder cost distributions. Similarly, in auction-like models (Guerre et al. 2000, Shakhgildyan 2022), nonparametric estimation of the underlying valuation structure often serves as a first step, providing the foundation for subsequent structural estimation.

Most dynamic structural estimation exercises rely on the Markov Perfect Equilibrium (MPE) of a dynamic game. However, when the model is calibrated to closely fit real-world data, the MPE typically lacks a closed-form solution. For example, Jiang et al. (2022) develop a finite-horizon dynamic game with discrete time, state, and action spaces to model creators’ strategic behavior in response to feedback during online logo design contests. Using data from 810 contests, they implement a two-step maximum likelihood estimation (MLE) procedure under the assumption of ex-ante homogeneous contests and players: the first step estimates action-specific state transition probabilities, while the second recovers utility parameters.

Specific to online crowdsourcing environments, Zhang et al. (2019) build a dynamic structural model to study how participants in crowdsourcing contests learn and choose contests over time. Their model shows that competing against superstars accelerates learning by providing more informative feedback. Using Bayesian updating and a hierarchical model, they estimate how individuals trade off monetary rewards, learning opportunities, and participation costs, based on data from Topcoder.

Most closely related to our work, Lemus and Marshall (2021) structurally estimate contest behavior using Kaggle data. They propose a finite-horizon dynamic game in which, after each submission, players decide whether to continue or exit, essentially solving an optimal stopping problem. Players can only make this decision immediately after completing a submission, which arrives at random intervals and incurs a privately drawn cost. Their estimation proceeds in two stages: the first uses maximum likelihood and the EM algorithm to estimate primitive parameters such as the distribution of player types and scores; the second applies a conditional choice probability

(CCP)-based estimator to identify the cost distribution. Our model differs in two key respects: (1) we treat submissions as the outcome of accumulated effort rather than as discrete costly actions; and (2) we explicitly model belief updating from noisy signals as a structural channel that shapes both behavior and identification, whereas their CCP-based approach relies on a coarser behavioral approximation.

The structure of the paper is as follows: Section 2 introduces our theoretical model of online crowdsourcing contests. Section 3 presents the Bayesian estimation framework. In Section 4, we describe the data-generating process and apply the Bayesian framework to synthetic data. Section 5 applies the framework to real-world Kaggle data and evaluates the estimated primitives using private-leaderboard path values excluded from the likelihood. A replication package (code and implementation details) will be provided with the paper.

2. The Model

Two players, i and j , compete for a prize $\theta > 0$ in a contest. The winner receives the prize and the loser receives zero. The contest starts at time zero. At each time $t \geq 0$, player i chooses an effort level $q_{i,t}$ and incurs quadratic cost $C_i(q_{i,t}) = c_i q_{i,t}^2 / 2$, where lower c_i corresponds to higher ability. Let y_t denote the *output gap* between players i and j at time t , evolving as

$$dy_t = (q_{i,t} - q_{j,t})dt + \sigma dW_t \quad (1)$$

where W_t is a Brownian motion and $\sigma > 0$ measures the innovation risk. When σ is large and the relationship between effort and output is highly uncertain, luck becomes the dominant factor. Conversely, when σ is small and output closely reflects effort, the importance of effort is amplified.

The actual contest is equipped with a submission system in which participants submit solutions over time, with each submission updating both the hidden private-leaderboard output gap and its noisy public-leaderboard readout. For tractability, we approximate this irregular sequence of submission-triggered updates by a continuous-time public signal of the latent output gap y_t . The precision of this signal is governed by λ . The signal evolves according to

$$dZ_t = y_t dt + \frac{dB_t}{\sqrt{\lambda}} \quad (2)$$

where B_t is a standard Brownian motion independent of W_t , and λ controls signal precision. A larger λ implies less noise and more informative feedback; in the empirical mapping, this corresponds to a more informative public leaderboard signal about latent relative performance. This continuous-time signal is an idealized analogue of the stream of public leaderboard updates observed after submissions.

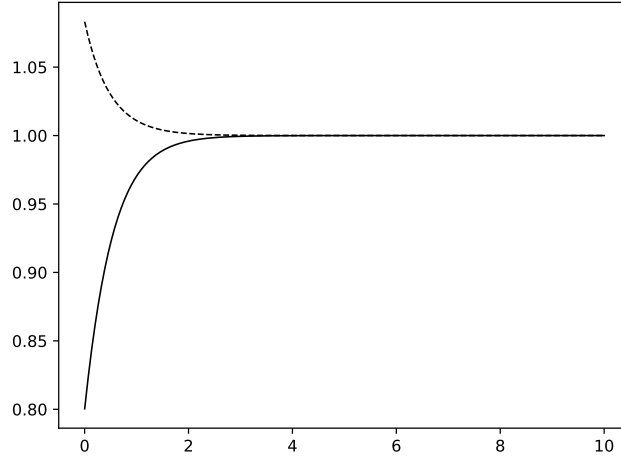


Figure 2 The evolution of S_t in time t , given that $\lambda = 1$ and $\sigma = 1$.

Note: The plot shows the conditional variance S_t of the filtering error (i.e., the uncertainty about the true performance gap). It converges quickly from any initial value to its steady state $\bar{S} = \sigma/\sqrt{\lambda}$. The figure illustrates the convergence of S_t to its steady state.

The information set of both players at time $t \geq 0$ is $I_t \equiv \{Z_s : 0 \leq s \leq t\}$. Player i estimates the unknown output gap y_t based on I_t . Let $\tilde{y}_t \equiv E(y_t | I_t)$ denote the filtered mean output gap, and let $S_t \equiv E[(\tilde{y}_t - y_t)^2 | I_t]$ denote the conditional mean-squared filtering error. Under standard linear-Gaussian assumptions, the Kalman-Bucy filter (Chapter 1.2 of [Bensoussan 1992](#); see also [Frogerais et al. 2012](#), [Barrau and Bonnabel 2017](#)) implies

$$d\tilde{y}_t = (q_{i,t} - q_{j,t})dt + \lambda S_t(dZ_t - \tilde{y}_t dt) \quad (3)$$

$$\frac{dS_t}{dt} = \sigma^2 - \lambda S_t^2 \quad (4)$$

Hence, the conditional distribution satisfies $y_t | I_t \sim \mathcal{N}(\tilde{y}_t, S_t)$, so the posterior state is fully characterized by $(\tilde{y}_t, S_t)$. The steady-state value associated with equation (4) is $\bar{S} \equiv \sigma/\sqrt{\lambda}$. For analytical tractability, we set $S_0 = \bar{S}$, so that $S_t \equiv \bar{S}$ for all t . Fixing S_t at its steady-state value leaves $\tilde{y}_t$ as the only performance state and makes a closed-form equilibrium possible. However, the model can no longer capture how filtering uncertainty changes over the course of a contest or how such changes affect effort. The general solution of equation (4) is reported in [Appendix B.1](#). [Figure 2](#) illustrates convergence of S_t to $\bar{S}$ from arbitrary initial conditions.

Following [Ryvkin \(2022\)](#), we consider a dynamic contest with a fixed deadline. Suppose the contest is terminated when time $t = T > 0$. Under the steady-state restriction $S_t \equiv \bar{S}$, the payoff-relevant state is fully characterized by $(\tilde{y}_t, t)$. Let $\tilde{y}_T$ denote player i 's perceived terminal performance advantage over player j . To retain the tractable tug-of-war structure of [Ryvkin \(2022\)](#), we specify terminal payoffs based on the filtered mean: player i receives the prize if $\tilde{y}_T > 0$, whereas player j receives it if $\tilde{y}_T < 0$. This threshold rule replaces a terminal expected payoff based on the

posterior winning probability, such as $\theta \Pr(y_T > 0 \mid I_T)$. At any time $0 \leq t < T$, player i optimizes her effort level $q_{i,\tau}$ in the remaining contest period $\tau \in [t, T)$ according to the following optimization problem,

$$V^i(\tilde{y}_t, t; q_{j,t}, \Theta_i) = \max_{\{q_{i,\tau}\}_{\tau=t}^T} \mathbb{E} \left(\theta \cdot \mathbf{1}_{\tilde{y}_T > 0} - \int_t^T C_i(q_{i,\tau}) d\tau \mid I_t \right) \quad (5)$$

where $\Theta_i \equiv \{\theta, \lambda, \sigma, c_i\}$, subject to constraints (3), (4) and $q_{i,\tau} \geq 0$ for all $\tau \in [t, T)$. The optimization problem for player j is symmetric to that of player i ; with player labels and cost parameters relabeled, we write $V^j(\tilde{y}_t, t) = V^i(-\tilde{y}_t, t)$. The corresponding Hamilton-Jacobi-Bellman (HJB) equation for player i is

$$0 = \max_{q_{i,t} \geq 0} \left[-\frac{c_i q_{i,t}^2}{2} + V_y^i \cdot (q_{i,t} - q_{j,t}) + V_t^i + \frac{V_{yy}^i}{2} \lambda \bar{S}^2 \right].$$

By definition, we have $\lambda \bar{S}^2 = \sigma^2$. Assuming an interior solution, substituting the first-order conditions $q_{i,t} = V_y^i / c_i$ and $q_{j,t} = -V_y^j / c_j$ yields the system

$$\begin{aligned} \frac{1}{2c_i} (V_y^i)^2 + \frac{1}{c_j} V_y^i V_y^j + V_t^i + V_{yy}^i \frac{\sigma^2}{2} &= 0 \\ \frac{1}{2c_j} (V_y^j)^2 + \frac{1}{c_i} V_y^j V_y^i + V_t^j + V_{yy}^j \frac{\sigma^2}{2} &= 0 \end{aligned}$$

subject to boundary conditions $V^i(-\infty, t) = 0$, $V^i(+\infty, t) = \theta$, $V^j(-\infty, t) = \theta$, $V^j(+\infty, t) = 0$, $V^i(\tilde{y}_T, T) = \theta \mathbf{1}_{\{\tilde{y}_T > 0\}}$ and $V^j(\tilde{y}_T, T) = \theta \mathbf{1}_{\{\tilde{y}_T < 0\}}$.

The Nash equilibrium is summarized in the following theorem.

THEOREM 1 (Ryvkin 2022). *Under the terminal-payoff rule above, the game takes the form of a fixed-deadline tug-of-war contest. In the Markov perfect equilibrium, the players' efforts in state $m_{i(j)}(\tilde{y}_t, t) : \mathbb{R} \times [0, T) \rightarrow \mathbb{R}_+$ are given by*

$$m_{i(j)}(\tilde{y}_t, t) = \frac{e^{-z^2/2}}{\sqrt{2\pi\sigma^2(T-t)}} \cdot \frac{\sigma^2}{2} [\gamma(\rho_i) + \gamma(\rho_j)] [1 - \rho(z)^2] [1 \pm \rho(z)] \quad (6)$$

where $z = \tilde{y}_t / (\sigma\sqrt{T-t})$, $\rho(z) = \gamma^{-1}(\Phi(z) [\gamma(\rho_i) + \gamma(\rho_j)] - \gamma(\rho_j))$ and

$$\gamma(u) = \frac{u}{1-u^2} + \frac{1}{2} \ln \frac{1+u}{1-u}, \quad u \in (-1, 1) \quad (7)$$

In the compact notation $m_{i(j)}$, the plus sign in $[1 \pm \rho(z)]$ corresponds to player i , and the minus sign corresponds to player j .

$$\rho_i = \frac{e^{w_i} + e^{-w_j} - 2}{e^{w_i} - e^{-w_j}}, \quad \rho_j = \frac{e^{w_j} + e^{-w_i} - 2}{e^{w_j} - e^{-w_i}}, \quad w_{i(j)} = \frac{\theta}{\sigma^2 c_{i(j)}}.$$

Moreover, the total expected effort at time t , defined by $M(\tilde{y}_t, t) := \mathbb{E} \left[\int_t^T m_i(\tilde{y}_s, s) + m_j(\tilde{y}_s, s) ds \right]$, is given by $M(\tilde{y}_t, t) = \sigma\sqrt{T-t}q(z)$, where $q(z)$ is determined by the ordinary differential equation

$$\begin{aligned} q''(z) + \left[4A_+(z)e^{-z^2/2} + z \right] q'(z) - q(z) + 4A_-(z)e^{-z^2/2} &= 0 \\ A_-(z) &= \frac{\gamma(\rho_i) + \gamma(\rho_j)}{2\sqrt{2\pi}} [1 - \rho(z)^2], \quad A_+(z) = A_-(z)\rho(z) \end{aligned} \quad (8)$$

subject to the boundary conditions $q(z) \rightarrow 0$ as $z \rightarrow \pm\infty$. Specifically, when $c_i = c_j$, we have $w_i = w_j = w$ and the total expected effort is reduced to

$$M(\tilde{y}_t, t) = \frac{2w\sigma\sqrt{T-t}}{\sqrt{2\pi}} e^{-z^2/2} + o(w) \quad (9)$$

REMARK 1. The approximation is accurate when $w = \theta/(\sigma^2 c)$ is small, i.e., when uncertainty σ is high relative to the prize-cost ratio θ/c . This is the regime where the first-order expansion in w is most informative.

In equation (6), w_i and w_j summarize prize incentives relative to innovation risk, and ρ_i and ρ_j are normalized transformations of these indices. The equilibrium effort $m_{i(j)}$ can be decomposed into two components: a normal density term $\phi(\tilde{y}_t; 0, \sigma^2(T-t))$ and an amplitude term $K_{i(j)}(z; \theta, \sigma, c_i, c_j) := \frac{\sigma^2}{2} [\gamma(\rho_i) + \gamma(\rho_j)] [1 - \rho(z)^2] [1 \pm \rho(z)]$. The density term governs how competitive intensity is distributed over the state space, whereas the amplitude term governs the asymmetry of effort across players.

This representation yields three observations useful for the empirical analysis. First, the parameters θ , c_i , and c_j enter the equilibrium effort expression through the amplitude term, so ability and prize incentives primarily affect the level and asymmetry of effort. Second, σ affects both the dispersion of the density term and the amplitude term, so its role is not reducible to a simple scale effect. Third, as the remaining time $T - t$ shrinks, effort becomes increasingly concentrated near the pivotal state.

The equilibrium characterization is inherited from Ryvkin (2022) after the filtered-state reduction to $(\tilde{y}_t, t)$ under $S_t \equiv \bar{S}$. Additional properties of the equilibrium are discussed in Ryvkin (2022). Here, we restrict attention to a brief design discussion under the maintained steady-state specification.

2.1. Implications on the Contest Design

In online crowdsourcing contests, designers typically control key decision variables such as the prize amount θ , contest duration T , and the informativeness of feedback, represented here by signal precision λ . We abstract from more complex prize structures in this study. A key concern is that these variables may act as substitutes, so the optimal choice of one design variable may alter the value of changing another.

Under the perceived-state terminal payoff and the steady-state filtering assumption $S_t = \bar{S}$, signal precision λ does not enter the closed-form equilibrium effort policy or total expected effort $M(\tilde{y}_t, t)$. This neutrality follows from the steady-state relation $\lambda\bar{S}^2 = \sigma^2$, which causes signal precision to drop out of the effort policy conditional on the perceived state. The result is specific to the simplified specification: if terminal payoffs depended on posterior winning probabilities or filtering uncertainty varied over time, λ could affect equilibrium effort.

Next, we turn to the other two design variables. When the two contestants are of similar strength, i.e., $c_i = c_j = c$, the total expected effort admits a particularly simple approximation, as shown in equation (9). Suppose, for this design discussion, that the contest designer's reduced-form objective takes the form $\beta^T M(\tilde{y}_0, 0)^\alpha - \theta$, where $0 < \alpha < 1$ and $0 < \beta \leq 1$. The parameter $\beta \in (0, 1]$ captures the time cost or discounting associated with longer contests. The parameter $\alpha \in (0, 1)$ captures diminishing returns from aggregate contest effort to the designer's value. Although higher effort increases the chance of obtaining a better solution, the marginal value of additional effort is likely to decline once the contest has already attracted substantial search activity. Thus, M^α should be interpreted as a reduced-form mapping from total expected effort to the designer's effective value from the contest. This concavity also ensures that the designer's benefit from increasing the prize does not grow without bound relative to the linear prize cost.

Let $\bar{M}(\tilde{y}_0, 0)$ denote the leading-order approximation to total effort obtained from equation (9) under the small- w approximation, $\bar{M}(\tilde{y}_0, 0) = \frac{2w\sigma\sqrt{T}}{\sqrt{2\pi}} e^{-z^2/2}$, where $w = \theta/(\sigma^2 c)$ and $z = \tilde{y}_0/(\sigma\sqrt{T})$. This approximation turns the designer's objective into an explicit function of (θ, T) , making it possible to derive the comparative statics below.

PROPOSITION 1. *Consider the symmetric benchmark with $c_i = c_j = c$, and fix the initial perceived gap $\tilde{y}_0$, the innovation risk σ , and the effort cost parameter c . Under the leading-order approximation in equation (9), let $\bar{M}(\tilde{y}_0, 0)$ denote the approximated total expected effort. Define the corresponding approximated designer objective as $\bar{\Pi}(\theta, T, \lambda) = \beta^T \bar{M}(\tilde{y}_0, 0)^\alpha - \theta$, where $0 < \alpha < 1$ and $0 < \beta \leq 1$. Then the following statements hold:*

(1) *The approximated total effort $\bar{M}(\tilde{y}_0, 0)$ is increasing in the prize amount θ and increasing in the contest duration T .*

(2) *For any fixed $T > 0$, the approximated designer objective $\bar{\Pi}$ is single-peaked in the prize amount θ . For any fixed $\theta > 0$, if $0 < \beta < 1$, $\bar{\Pi}$ is single-peaked in the contest duration T ; if $\beta = 1$, $\bar{\Pi}$ is increasing in T .*

(3) *If the designer jointly optimizes over (θ, T) on $\theta > 0$ and $T > 0$, then for $0 < \beta < 1$, $\bar{\Pi}$ has at most one interior optimum. If $\beta = 1$, $\bar{\Pi}$ is increasing in T , so the problem has no finite optimal duration.*

Moreover, within this steady-state filtering approximation, both $\bar{M}(\tilde{y}_0, 0)$ and $\bar{\Pi}(\theta, T, \lambda)$ are neutral with respect to the signal precision λ .

Although λ is neutral with respect to the total-effort design object above, it remains a structural information primitive that can be identified in the likelihood. In the estimation framework below, λ governs the Kalman correction in the perceived-state recursion and the signal-noise component of the public leaderboard likelihood.

3. Estimation Framework

This section presents the data-generating process and the Bayesian estimation procedure. We first connect the theoretical model to the observable contest data and then describe the likelihood construction and prior specification.

3.1. Data-Generating Process

For each Kaggle contest, the observable data from Meta-Kaggle can be grouped into three components.

The first component contains contest-level institutional features: contest duration, prize structure, and information disclosure rules. To match the two-player winner-take-all structure of the model, we restrict the empirical application to contests where a dominant top rivalry is plausible and monetary prize incentives can be mapped to an effective top-two winner-gain structure.

The second component records submission events of player i , denoted by $\{\hat{t}_k^i\}_{k=1}^{N_i}$, where N_i is the total number of submissions. Conditional on latent effort paths, we model the submissions of players i and j as two inhomogeneous Poisson processes with intensities $\tau_i(t)$ and $\tau_j(t)$. These processes are conditionally independent given the latent effort paths. For any interval $\mathcal{S} \subset \mathcal{T}$, the expected number of submissions by player i is $\int_{\mathcal{S}} \tau_i(s) ds$. This proportionality condition should be interpreted as a measurement bridge from latent equilibrium effort to observed actions, rather than as a theorem of the game. For tractability, we assume

$$\tau_i(t) = r \cdot m_i(\tilde{y}_t, t) \quad (10)$$

where $r > 0$ scales latent effort into observed submission intensity. In estimation, r is treated as an auxiliary scale parameter rather than a contest primitive: it absorbs the unit conversion between model effort and submission rates, with its range disciplined by the prior support specified below.⁷

The third component contains *public* and *private* leaderboard series, denoted by $\hat{x}_t^{i(j)}$ and $x_t^{i(j)}$. In Kaggle contests, public scores are computed from a released subset, while private scores are computed from withheld data and revealed only after the contest ends. We define the latent contemporaneous gap as $y_t := x_t^i - x_t^j$ and the displayed public gap as $\hat{y}_t := \hat{x}_t^i - \hat{x}_t^j$, where each displayed score is carried forward from that contestant's most recent submission. The public gap is the empirical counterpart of the noisy signal in (2), observed through discrete leaderboard updates. We write this empirical readout as

$$dZ_t = \hat{y}_t dt, \quad (11)$$

⁷ Because r enters the submission-intensity equation rather than the public-signal variance or Brownian innovation covariance directly, its main identification interaction is expected to be with the absolute effort-cost scale. A synthetic scale-sensitivity check in the replication package re-estimates the baseline synthetic contest while treating r as unknown. The results are consistent with this interpretation, although they do not rule out indirect finite-sample interactions with σ or λ .

which should be interpreted as an observation convention linking the public leaderboard gap to the signal path, not as a second structural definition of Z_t . Combining (2) and (11) gives the formal white-noise representation

$$\hat{y}_t = y_t + \frac{1}{\sqrt{\lambda}} \xi_t, \quad \xi_t \equiv \frac{dB_t}{dt}, \quad (12)$$

where ξ_t denotes Gaussian white noise in the generalized-process sense. Because white noise is not pointwise observed, equation (12) is not a pointwise sampling equation. In the empirical data, the public leaderboard is sampled only at submission-triggered update times. At an empirical update time t_k , averaging the white-noise representation over the observed readout interval $h_k = t_k - t_{k-1}$ yields a finite-variance noise term $\varepsilon_k \sim \mathcal{N}(0, 1/(\lambda h_k))$. Approximating the corresponding readout of the latent gap by its endpoint value and substituting the latent state dynamics in (1) gives

$$\hat{y}_{t_k} \approx y_0 + \int_0^{t_k} [m_i(\tilde{y}_s, s) - m_j(\tilde{y}_s, s)] ds + \sigma W_{t_k} + \varepsilon_k, \quad (13)$$

for update times t_k . Here, σW_{t_k} captures accumulated innovation noise, while ε_k captures the public-signal noise associated with the leaderboard update. Thus, larger λ implies more precise public feedback. In the empirical implementation, we normalize $y_0 = 0$.

The theoretical model uses a continuous public signal and therefore admits the continuous-time filtering equation in (3). The empirical leaderboard, by contrast, is observed only at asynchronous submission times. A fully coherent empirical implementation would therefore require a hybrid filter with continuous prediction between submissions and discrete measurement updates when new scores arrive.

To retain the tractable one-dimensional state representation used in the equilibrium analysis, we employ a continuous-update approximation in the empirical implementation. Specifically, we interpolate the most recently observed public-gap process and evolve the perceived gap according to

$$d\tilde{y}_t = [m_i(\tilde{y}_t, t) - m_j(\tilde{y}_t, t)] dt + \sqrt{\lambda} \sigma (\hat{y}_t - \tilde{y}_t) dt \quad (14)$$

with initial condition $\tilde{y}_0 = \mu_0$. This approximation treats the carried-forward leaderboard gap as a continuous proxy for the unavailable signal path. It may overstate learning during long intervals without submissions, so the resulting estimates should be interpreted as conditional on this continuous-update approximation rather than as estimates from an exact discrete-observation filter.

We denote the set of estimated parameters by $\Theta := \{c_i, c_j, \sigma, \lambda, \mu_0, r\}$. Here, μ_0 primarily captures the initial perceived gap and is treated as a nuisance initial-condition parameter rather than a structural design primitive. The scale parameter r is estimated as an auxiliary measurement parameter that maps latent effort into submission intensity. Once Θ is specified, equations (1), (13), and (14) define the working empirical system, while realized trajectories remain stochastic.

3.2. Modeling Considerations

This subsection discusses three practical modeling concerns that are important for interpretation: strategic submission timing, the two-player approximation, and the normality assumption. The goal is to clarify the scope of the baseline framework and connect each concern to the robustness analyses that follow.

Strategic submission timing. Beyond the uncertainty caused by partial data disclosure, an additional source of signal noise stems from the timing of leaderboard updates: rankings are only refreshed upon new submissions, rendering the leaderboard uninformative during periods of inactivity. This lag can bias inference if participants deliberately time or withhold submissions.

Our baseline observation equation assumes submissions are driven by latent effort and does not explicitly model strategic withholding. Strategic submission timing is an important concern in the literature. For example, [Lemus and Marshall \(2021\)](#) adopt a similar simplification by excluding such behavior from their model. Our estimates should therefore be interpreted for contests where active updating behavior dominates withholding behavior. As a partial mitigation, we require a minimum number of submissions when selecting representative contestants (Section 5.1), which reduces cases where sparse strategic timing dominates observed dynamics.

Selection of Two Players. Another concern is the two-player approximation. Real contests involve many participants, so reducing the environment to two players changes strategic externalities and information structure. Our interpretation is that the model isolates the dominant top rivalry in contests where competition is concentrated near the frontier. In Section 6.1, we conduct a series of robustness checks. By varying the focal-pair definition, we assess whether the validation results depend on a particular choice of representative rivalry.

Gaussian approximation. The model relies on Gaussian signal and innovation shocks. Mapping bounded leaderboard scores into the model’s state space therefore requires a maintained distributional approximation. The logit transformation is adopted for tractability but does not ensure Gaussianity. Alternative values of its multiplicative scale parameter a assess sensitivity to units, not sensitivity to distributional shape. We therefore interpret the Gaussian specification as a maintained modeling assumption rather than as an empirically validated distributional result.

3.3. Bayesian Estimation Framework

Our objective is to develop a Bayesian framework for inferring the unknown parameter set $\Theta := \{c_i, c_j, \sigma, \lambda, \mu_0, r\}$ using the data introduced above.

Conditional on the selected focal pair, the estimation likelihood uses only contest-phase submission timing and public leaderboard updates.

For numerical integration of the effort and filtering equations, we use a uniform computational grid of length Δ_{num} . This numerical step size does not determine the observation-noise variance. By

definition, the likelihood function corresponding to the submission times of contestant i , $\{t_k^i\}_{k=1}^{N_i}$, is given as follows (a symmetric formulation applies to contestant j):

$$p\left(\{\hat{t}_k^i\}_{k=1}^{N_i} | \tau_i, \Theta\right) = \exp\left\{-\int_{s \in \mathcal{T}} \tau_i(s) ds\right\} \prod_{k=1}^{N_i} \tau_i(\hat{t}_k^i) \quad (15)$$

where τ_i is defined in (10). The integral term is evaluated on the computational grid.

Next, the public leaderboard gap $\hat{y}_t$ enters the likelihood only when either contestant submits and receives an update. We denote these update times by the ordered sequence $(t_1, t_2, \dots, t_N)$, where $N = N_i + N_j$. Each t_k is associated with an observed public gap $\hat{y}_{t_k}$. Let $h_k = t_k - t_{k-1}$, with $t_0 = 0$, denote the elapsed calendar time associated with the k th empirical readout. The quantities h_k are determined by observed submission timing and are distinct from the numerical integration step Δ_{num} .

To make the conditioning structure explicit, we evaluate the leaderboard likelihood sequentially. Let $\mathcal{F}_{k-1}$ denote the submission and public-leaderboard history strictly before update time t_k . Conditional on $\mathcal{F}_{k-1}$ and Θ , the empirical filtering recursion generates the one-step-ahead predictive distribution

$$\hat{y}_{t_k} | \mathcal{F}_{k-1}, \Theta \sim \mathcal{N}(\mu_{k|k-1}, V_{k|k-1}),$$

where $\mu_{k|k-1}$ and $V_{k|k-1}$ are obtained from the prediction step over $(t_{k-1}, t_k]$ before the update at t_k is incorporated. The leaderboard likelihood is therefore

$$p\left(\{\hat{y}_{t_k}\}_{k=1}^N | \Theta\right) = \prod_{k=1}^N \phi(\hat{y}_{t_k} | \mu_{k|k-1}, V_{k|k-1}). \quad (16)$$

The observation at t_k is incorporated into the perceived-state recursion only after its conditional density has been evaluated, so it affects subsequent predicted states and effort paths but not its own one-step-ahead predictive mean. More generally, the mean would include the latent initial gap y_0 ; in the empirical implementation we normalize $y_0 = 0$ by measuring subsequent gaps relative to the initial latent-gap origin. This normalization is distinct from μ_0 , which is the initial perceived gap $\tilde{y}_0$. Thus, μ_0 enters the likelihood indirectly through the filtering recursion in (14), which determines the predictive states, effort paths, and conditional moments used in each factor. Accordingly, the likelihood uses the public leaderboard values associated with submission updates and preserves the temporal ordering of information.

It is worth noting that, although the private leaderboard trajectory y_t is observable ex post in Meta-Kaggle, we deliberately exclude it from the estimation likelihood. In our framework, agents form beliefs about y_t based solely on $\hat{y}_t$, and their actions are driven by these beliefs. Conditioning on the realized values of y_t in estimation would effectively bypass the agent's informational constraints

and undermine the role of λ in shaping belief formation. More importantly, it would destroy the path-wise holdout design based on private-leaderboard values. Treating y_t as latent preserves the model’s behavioral interpretation and enables post-estimation path validation, while final private ranks remain part of focal-pair selection.

To evaluate (15) and (16), we recursively construct $\tilde{y}_t$ and implied efforts (m_i, m_j) from (6) and (14). More precisely, conditional on a parameter vector, perceived states and effort paths are constructed by forward filtering, using each public update only for subsequent states. However, evaluating $m_{i(j)}(\tilde{y}_t, t)$ via equation (6) requires numerically approximating the inverse function γ^{-1} , which poses challenges for the use of gradient-based Markov Chain Monte Carlo (MCMC) sampling methods, such as Hamiltonian Monte Carlo (HMC, Neal 1996, Neal 2011, Betancourt 2017) and the No-U-Turn Sampler (NUTS, Hoffman and Gelman 2014). Hence, we approximate this inverse function with an analytical form:

$$\gamma^{-1}(x) \approx \frac{2}{\pi} \arctan(a \cdot x) \quad (17)$$

where $a \approx 0.856$ is obtained by minimizing the infinity norm of the difference between the numerical inverse of $\gamma(\cdot)$ and the approximation $\frac{2}{\pi} \arctan(ax)$.⁸ Figure 3 compares the equilibrium effort function $m_i(\tilde{y}_t, t)$ derived from the approximate analytical form of γ^{-1} in equation (17) with that obtained from the numerically accurate solution. As illustrated, the approximation closely replicates the true function; the approximation error appears negligible relative to posterior uncertainty over the calibration range considered.

In the empirical application, we use truncated normal priors with explicit supports: $c_i, c_j \sim \mathcal{TN}(1, 5; [0.5, 5])$, $\sigma \sim \mathcal{TN}(1, 5; [0.5, 10])$, $\lambda \sim \mathcal{TN}(1, 5; [10^{-6}, \infty))$, $r \sim \mathcal{TN}(10, 5; [10^{-2}, 100])$, and $\mu_0 \sim \mathcal{N}(\hat{y}_0, 5^2)$. Thus, priors are weakly informative for $(c_i, c_j, \sigma, \lambda)$, while r is an auxiliary scale parameter and μ_0 is intentionally more informative to stabilize initialization.

The following lemma formalizes local identification of the structural parameters under the maintained specification, conditional on a fixed effort-to-submission scale r . Identification comes from complementary moments: (c_i, c_j, μ_0) jointly shape asymmetric effort, submission timing, and early perceived-state dynamics through the point-process likelihood and the public-gap mean path; σ loads on the off-diagonal covariance of public-gap observations over time; and λ loads on the diagonal signal-noise component. In the Bayesian implementation above, r is still estimated as an auxiliary measurement scale, but the structural identification statement below is conditional on that scale. The proof is provided in the Appendix.

LEMMA 1. *For fixed r , under the maintained structural specification in (6), (14), (15) and (16), the parameter vector $(c_i, c_j, \sigma, \lambda, \mu_0)$ is locally identified, provided that the realized public-history support satisfies the full-rank variation condition stated in the proof.*

⁸ Under the L_1 -norm, the parameter $a = 0.947$; under the L_2 -norm, the parameter $a = 0.895$.

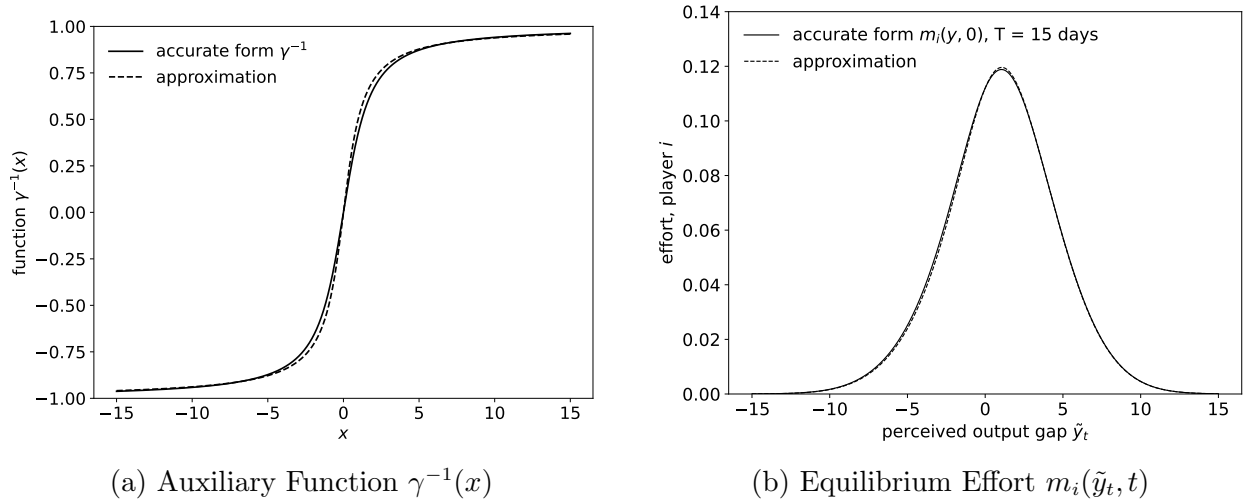


Figure 3 Comparison of the Accurate and Approximate Forms

Note: Panel (a) compares the exact inverse function $\gamma^{-1}(x)$ of the equilibrium component in (7) with its analytical approximation $\frac{2}{\pi} \arctan(ax)$ by (17), where $a \approx 0.856$ minimizes the infinity-norm approximation error. Panel (b) compares the resulting equilibrium effort $m_i(\tilde{y}_t, t)$ at $t = 0$ (with $T = 15$, $\theta = 1$, $c_i = c_j = 1$, $\sigma = 1$) obtained from the analytical approximation versus the exact numerical solution.

4. Synthetic Experiments

Before applying our estimation procedure to real-world contest data, we evaluate its potential on synthetically generated data. These synthetic exercises serve three purposes: illustrating the data-generating mechanism, assessing finite-sample parameter recovery, and clarifying which parameters are easier or harder to recover under realistic data conditions.

A central challenge in generating synthetic data lies in dynamically constructing a point process that conforms to an inhomogeneous Poisson process. We first generate candidate submission events according to a homogeneous Poisson process over the whole contest duration, using a fixed high intensity $\tau^* \geq \sup_t \tau_{i(j)}(t)$. Then, we apply the classical thinning procedure (Lewis and Shedler 1979) to determine whether each candidate event is accepted or not. In plain terms, Algorithm 1 generates candidate submissions, simulates latent and perceived states, accepts submissions via thinning, updates public leaderboard observations, and produces the synthetic sample used in the subsequent Bayesian recovery exercise.

As a baseline example, we consider a synthetic contest with a three-month horizon. The two participants differ slightly in effort costs, with $c_i = 1.2$ and $c_j = 1.5$. The remaining parameters are set to $\theta = 1$, $\sigma = 2.0$, $\lambda = 1.0$, and $r = 15$. Under this specification, we simulate the latent output gap, the public leaderboard path, the players' submission times, the perceived output gap, and the corresponding equilibrium effort paths.

Figure 4 shows one realization of the baseline synthetic contest. The blue line represents the latent output gap y_t , which can be interpreted as the *private* leaderboard, while the solid black line

Algorithm 1 Synthetic Data Simulation

Input: $\Delta, T, \theta, c_i, c_j, \sigma, \lambda, r, \tau^*, y_0, \mu_0$

Sample $\{s_k^{i(j)}\}_{k=1}^{N_{i(j)}}$ from Poisson process (τ^*) on $[0, T]$

Sample $\{u_k^{i(j)}\}_{k=1}^{N_{i(j)}}$ from uniform distribution on $[0, 1]$

Sample series of Brownian motions (W_t) and (B_t)

Initialize $\ell^{i(j)} = 0, \hat{y}_0 = 0, \tilde{y}_t = \mu_0$

for $t = 0, \Delta, \dots, T - \Delta$ **do**

$m_{i(j)}(\tilde{y}_t, t) \leftarrow (6), \tau^{i(j)}(t) \leftarrow (10)$ {Use $\theta, c_{i(j)}, \sigma, r, T$ }

$y_{t+\Delta} \leftarrow (1)$ {Use $m_i - m_j, \sigma, \Delta, W_{t+\Delta}$ }

$\tilde{y}_{t+\Delta} \leftarrow (14)$ {Use $m_i - m_j, \lambda, \sigma, \Delta$, and available $\hat{y}_t$ }

for $s_k^{i(j)} \in [t, t + \Delta)$ **do**

if $u_k^{i(j)} < \tau_{i(j)}(s_k^{i(j)})/\tau_{i(j)}^*$ **then**

$\hat{t}_\ell^{i(j)} \leftarrow s_k^{i(j)}; \ell^{i(j)} \leftarrow \ell^{i(j)} + 1$ {Accept the submission event}

$\hat{y}_{t+\Delta} \leftarrow$ discretized readout of (13) {Use $\sigma, \lambda, B_{t+\Delta}$; stored for subsequent filtering}

end if

end for

end for

Output: $(y_t), (\tilde{y}_t), (m_{i(j)}(\tilde{y}_t, t)), \{\hat{t}_k^{i(j)}\}_{k=1}^{N_{i(j)}}, (\hat{y}_{t_k})_{k=1}^{N_i+N_j}$

represents the *public* leaderboard $\hat{y}_t$, updated whenever either player submits. The dashed black line represents the perceived output gap $\tilde{y}_t$, inferred from the public leaderboard as in equation (14). The short vertical ticks mark submission times generated by the inhomogeneous Poisson processes. In this realization, player i submits 28 times and player j submits 18 times. Between submissions, filtering uncertainty would generally evolve over time; for tractability, however, the synthetic design follows the maintained steady-state specification used throughout the model.

The synthetic recovery exercises use simulation-specific priors, reported in the table notes, and estimate $(c_i, c_j, \sigma, \lambda, \mu_0)$ while fixing the auxiliary effort-to-submission scale r at its simulation value. When describing a parameter as weakly pinned down in these Bayesian recovery exercises, we use a simple posterior-concentration diagnostic: the posterior standard deviation remains close to its prior standard deviation, and the credible interval shows little shrinkage relative to the prior support. For μ_0 , the prior standard deviation is approximately one, so posterior standard deviations near one indicate limited learning from the realized synthetic data.

Table 1 presents the Bayesian inference results based on the synthetic data that we display in Figure 4. With limited data from a short contest, recovery is uneven across parameters. In this baseline case, c_i, c_j , and λ are reasonably recoverable, while σ is harder to recover and μ_0

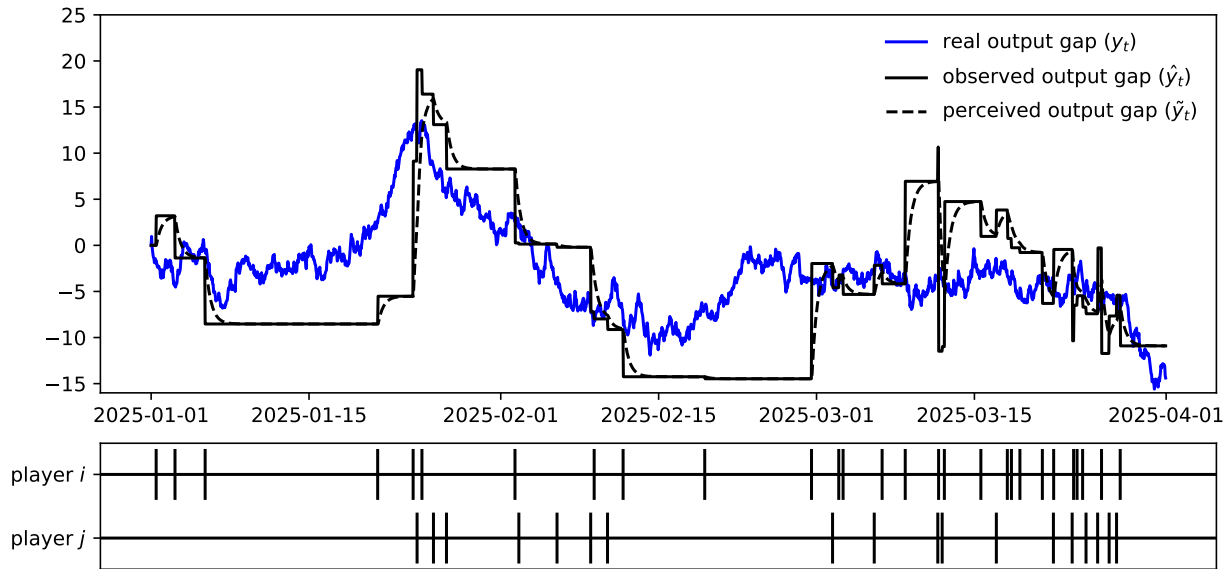


Figure 4 Synthetic Contest Data over a Three-Month Horizon

Note: The blue line represents the latent output gap y_t (private leaderboard), the solid black line represents the public leaderboard $\hat{y}_t$ (updated only upon submissions), and the dashed black line represents the perceived output gap $\tilde{y}_t$ inferred via Bayesian filtering. Vertical ticks mark submission times: ticks above the axis are player i 's submissions; ticks below are player j 's. Parameters: $\theta = 1.0$, $c_i = 1.2$, $c_j = 1.5$, $\sigma = 2.0$, $\lambda = 1.0$, $r = 15$, $T = 3$ months.

remains weakly pinned down (posterior standard deviation close to 1). This finite-sample pattern is informative and motivates the richer-data exercises below.

Table 1 Bayesian Estimates from Synthetic Data

Parameters (true val.)	c_i 1.2	c_j 1.5	σ 2.0	λ 1.0	μ_0 0.0
Posterior Mean	1.089	1.751	2.801	1.105	0.005
Posterior SD	0.258	0.476	0.503	0.299	0.990
Post. RMSD	0.281	0.538	0.945	0.317	0.990
95% Interval	[0.68, 1.68]	[1.01, 2.85]	[1.95, 3.93]	[0.62, 1.78]	[-1.92, 1.94]

Note: This table reports the posterior means, posterior standard deviations, posterior root mean squared deviations (Post. RMSD), and 95% credible intervals from Bayesian estimation using the synthetic contest data shown in Figure 4. In this single synthetic realization, players i and j submitted 28 and 18 times, respectively. Post. RMSD is defined as $\{(\bar{\theta} - \theta_0)^2 + s_{\bar{\theta}}^2\}^{1/2}$, where θ_0 is the true simulation value, $\bar{\theta}$ is the posterior mean, and $s_{\bar{\theta}}$ is the posterior standard deviation. It is a posterior recovery diagnostic for this realized synthetic sample, not a frequentist repeated-sampling quantity. The priors are $c_i, c_j \sim \mathcal{TN}(0.5, 5; [0.1, 5])$, $\sigma \sim \mathcal{TN}(1, 5; [0.5, 10])$, $\lambda \sim \mathcal{TN}(1, 5; [10^{-6}, 100])$, and $\mu_0 \sim \mathcal{TN}(0, 1; [-20, 20])$; r is fixed at its simulation value. The 95% credible interval reports the central posterior interval containing 95% of the posterior mass.

As an auxiliary scale-sensitivity check, we re-estimate the same baseline synthetic contest while treating the effort-to-submission scale r as an unknown parameter rather than fixing it at its simulation value. Table 2 reports the resulting posterior diagnostic. The main effect of estimating

r is to widen uncertainty about the absolute cost scale: posterior correlations between r and (c_i, c_j) are 0.709 and 0.656, respectively. By contrast, the corresponding posterior correlations with σ and λ are 0.055 and 0.009. Thus, in this synthetic exercise, adding r mainly affects absolute effort-cost scaling, while the innovation-risk and signal-precision primitives remain comparatively stable.

Table 2 Synthetic Scale-Sensitivity Check with Estimated r

Parameters (true val.)	c_i	c_j	σ	λ	μ_0	r
	1.2	1.5	2.0	1.0	0.0	15.0
Posterior Mean	1.294	1.927	2.786	1.134	0.004	18.439
Posterior SD	0.402	0.595	0.483	0.307	0.994	4.569
Post. RMSD	0.413	0.732	0.922	0.335	0.994	5.719
95% Interval	[0.64, 2.19]	[0.90, 3.17]	[1.94, 3.83]	[0.65, 1.84]	[-1.95, 1.95]	[11.09, 28.89]

Note: This table repeats the baseline synthetic exercise in Table 1 but estimates the auxiliary effort-to-submission scale r jointly with $(c_i, c_j, \sigma, \lambda, \mu_0)$. The prior for r is centered at its simulation value, $r \sim \mathcal{TN}(15, 5; [10^{-6}, 1000])$. The table is intended as a scale-sensitivity diagnostic for the baseline synthetic realization.

We next examine how recovery changes with richer data.

4.1. Parameter Recovery Under Richer Data

This subsection provides simulation-based recovery diagnostics under two richer-data regimes: pooling replicated contests under fixed structural parameters, and observing longer contests with more submissions. The goal is practical rather than asymptotic or theorem-based: to assess whether recovery improves in sensible ways as information accumulates.

Replications. To begin, we investigate recovery in a fixed-design information-scaling exercise. Starting from the baseline synthetic contest, we replicate the same contest design under identical structural parameters and pool the resulting panels for Bayesian inference. The submission counts are held fixed at the baseline realization in each replicated panel, so the exercise should not be read as an unconditional simulation of the Poisson count distribution. Instead, it asks whether posterior recovery improves when the likelihood is given more observations from the same underlying information environment.

The experimental results are presented in Table 3. We consider pools of 10 and 20 replicated panels. Compared with Table 1, posterior RMSD and posterior uncertainty decline for c_i , c_j , σ , and λ , showing that the likelihood contains meaningful identifying content that compounds as more contests are added. In contrast, μ_0 shows little posterior concentration even with pooled contests.

Long-Horizon Contest. We then turn to a single-contest regime with a longer time horizon. Rather than increasing the number of trajectories, we examine how the accumulation of submissions over a longer period improves inference accuracy.

Table 3 Bayesian Estimates from Pooled Synthetic Data

Parameters (true val.)	c_i	c_j	σ	λ	μ_0
	1.2	1.5	2.0	1.0	0.0
Pool of 10 Contests					
Posterior Mean	1.254	1.686	1.912	1.053	-0.009
Posterior SD	0.087	0.128	0.095	0.090	0.975
Post. RMSD	0.102	0.225	0.130	0.104	0.975
Pool of 20 Contests					
Posterior Mean	1.204	1.639	1.981	0.988	0.008
Posterior SD	0.060	0.089	0.068	0.060	1.003
Post. RMSD	0.060	0.165	0.071	0.062	1.003

Note: This table reports the posterior means, posterior standard deviations, and posterior root mean squared deviations (Post. RMSD) from Bayesian estimation using *pooled* synthetic panels under identical structural parameters. The pooled exercise fixes the realized submission counts from the baseline synthetic contest in each panel; hence the totals are exactly proportional to the single-contest counts (28 submissions by player i and 18 by player j). This design isolates how posterior recovery changes with the amount of information, rather than the additional sampling variation in total Poisson counts. The priors are $c_i, c_j \sim \mathcal{TN}(1, 5; [0.5, 5])$, $\sigma \sim \mathcal{TN}(1, 5; [0.5, 10])$, $\lambda \sim \mathcal{TN}(1, 5; [10^{-6}, 10])$, and $\mu_0 \sim \mathcal{TN}(0, 1; [-20, 20])$; r is fixed at its simulation value. Post. RMSD is defined as $\{(\bar{\theta} - \theta_0)^2 + s_{\bar{\theta}}^2\}^{1/2}$ and is computed from the posterior distribution conditional on the realized synthetic sample.

When generating long-horizon contest data, it is important to appropriately lower the effort costs c_i and c_j to ensure sustained effort from the contestants. This is because effort levels are influenced by the remaining time until the deadline: according to equation (6), the further away the deadline, the lower the equilibrium effort. Hence, this exercise is not a pure horizon comparison: we jointly vary horizon and effective submission intensity to keep the synthetic environment informative.

Table 4 shows that posterior recovery improves as within-contest information increases, although this improvement reflects both a longer horizon and the lower effort costs used to maintain active submission behavior.

Taken together, the synthetic exercises indicate a consistent recoverability ranking: c_i , c_j , and λ are comparatively well recovered; σ is harder in thinner samples but improves with richer data; and μ_0 remains weakly learned according to the posterior-concentration criterion above.

5. Empirical Application

In this section, we apply the model to real Kaggle contests and assess whether structurally recovered contest primitives align with private-leaderboard-based benchmarks.

We select representative Kaggle competitions from Meta-Kaggle and identify two focal participants i and j in each contest based on submission frequency, activity duration, and final ranking. Using their submission records and public leaderboard trajectories, we construct the observed data $\{t_k^i\}_{k=1}^{N_i}$, $\{t_k^j\}_{k=1}^{N_j}$ and $\{\hat{y}_{t_k}\}_{k=1}^{N_i+N_j}$ for contest-level Bayesian estimation under a unified prior specification.

Table 4 Bayesian Estimates from Long Term Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0
Contest of 6 Months					
True Value	0.6	0.75	2.0	1.0	0.0
Posterior Mean	0.806	0.936	1.685	1.116	0.011
Posterior SD	0.112	0.142	0.148	0.171	0.986
Post. RMSD	0.234	0.234	0.348	0.207	0.986
Contest of 12 Months					
True Value	0.3	0.375	2.0	1.0	0.0
Posterior Mean	0.333	0.436	2.067	1.101	0.012
Posterior SD	0.027	0.046	0.097	0.138	0.994
Post. RMSD	0.043	0.076	0.118	0.171	0.994

Note: This table reports the posterior means, posterior standard deviations, and posterior root mean squared deviations (Post. RMSD) from Bayesian estimation using synthetic contest data with longer horizons (6 and 12 months). In the 6-month contest, players i and j submitted 59 and 51 times, respectively; in the 12-month contest, they submitted 100 and 97 times, respectively. To sustain active submission behavior over longer horizons, effort costs c_i and c_j are reduced proportionally (true values: $c_i = 0.6$, $c_j = 0.75$ for 6 months; $c_i = 0.3$, $c_j = 0.375$ for 12 months). The priors are the same as in the baseline single-contest exercise: $c_i, c_j \sim \mathcal{TN}(0.5, 5; [0.1, 5])$, $\sigma \sim \mathcal{TN}(1, 5; [0.5, 10])$, $\lambda \sim \mathcal{TN}(1, 5; [10^{-6}, 100])$, and $\mu_0 \sim \mathcal{TN}(0, 1; [-20, 20])$; r is fixed at its simulation value. Post. RMSD is defined as $\{(\bar{\theta} - \theta_0)^2 + s_{\bar{\theta}}^2\}^{1/2}$ and is computed from the posterior distribution conditional on the realized synthetic sample.

The estimation likelihood uses only contest-phase submission timing and public leaderboard trajectories. Private-leaderboard paths and score values are excluded from the likelihood and are used to construct post-estimation validation targets. Final private ranks, however, enter the focal-pair selection rule, so the resulting design should be interpreted as path-wise holdout validation rather than complete sample-selection independence. Specifically, we compare structural contest-level estimates of λ and σ against benchmark quantities constructed from private-leaderboard path values excluded from the likelihood. By the private-leaderboard path, we mean the ex post sequence of private scores associated with the contestants' historical submissions, reconstructed after the contest; it is not a path observed by contestants in real time. Figure 5 summarizes the data-model structure and validation design.

5.1. Preparing Data for Structural Estimation

To empirically implement the structural estimation framework developed in the previous sections using Kaggle data, the first step is to map the observed data onto the corresponding variables defined by the model. This subsection details the procedures for selecting competitions, identifying key participants, and incorporating contest-specific information such as prize structures to make the data compatible with the model's structural assumptions.

As a first selection criterion, we focus on monetary prize structure. We restrict attention to Kaggle competitions offering one, two, or three monetary prizes denominated in U.S. dollars. Competitions with a single prize naturally conform to the winner-take-all incentive structure. For competitions

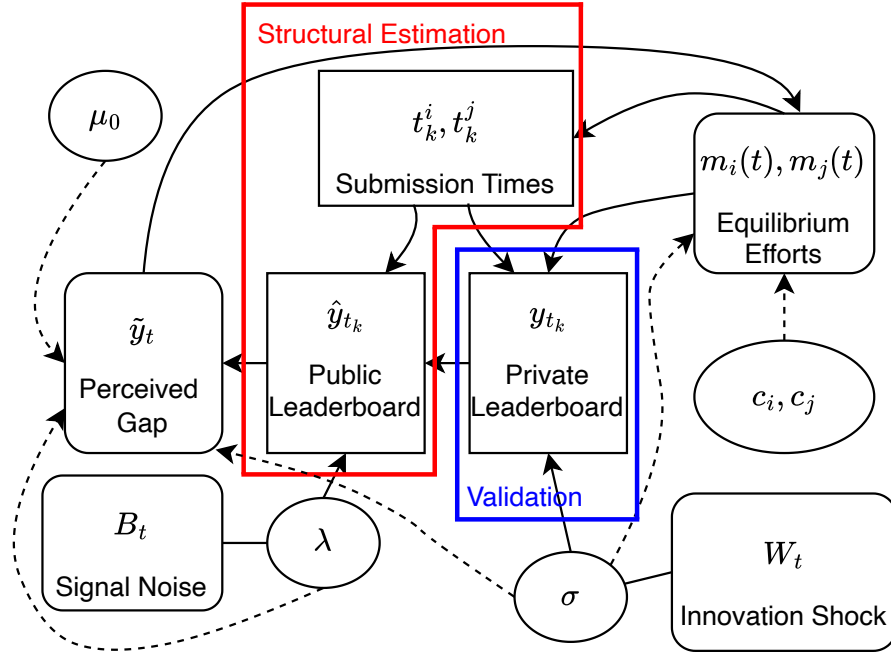


Figure 5 Meta-Kaggle Dataset and the Model Structure

Note: This figure shows the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles represent observed data, rounded rectangles represent latent states, and ellipses represent contest parameters. Solid lines depict temporal relationships among time series variables and state variables, while dashed arrows show the influence of contest parameters on the data-generating process. The private leaderboard y_t is not observed by contestants during the contest. Its path values are excluded from the likelihood and used for post-estimation validation, while final private ranks enter focal-pair selection.

with two or three prize tiers, we align the empirical setting with the model by treating the prize difference between the top two participants as the effective gain from winning. Because non-winning participants may still receive symbolic recognition such as medals, focusing on the top-two prize gap helps mitigate confounding incentives and brings the empirical setting closer to the model's winner-take-all benchmark.

The second criterion requires two highly active, top-ranked contestants who engage in sustained competition during the contest. We identify candidate focal contestants using final private leaderboard ranks, beginning from the top of the ranking. This choice aligns the two-player approximation with rivalry among contestants who perform best under the contest's final evaluation criterion. Private path values do not enter the estimation likelihood, but final private ranks determine the focal pair; the validation exercises are therefore path-wise holdout diagnostics conditional on ex post pair selection. To ensure sustained overlapping activity and adequate data for estimation, we impose two conditions: each contestant must submit at least five times, and the pair must satisfy the temporal-overlap criterion used in sample construction. These criteria help ensure that the focal-pair data contain enough submission activity and temporal overlap for structural estimation.

While analytically convenient, restricting attention to two participants may limit representativeness, oversimplify the strategic environment, and introduce selection concerns via *ex post* rankings. We therefore interpret this as a disciplined scope choice: a two-player structural approximation of the dominant rivalry near the top of contests with concentrated competition, frequent submissions, and overlapping activity windows.

The third criterion for contest selection concerns the format in which leaderboard scores are presented. Since Kaggle competitions are based on diverse real-world tasks, the scoring metrics vary accordingly. Some are based on raw scores (positive real numbers), while others are based on accuracy measures (expressed as percentages or decimals between 0 and 1). Ignoring these differences would lead to inconsistencies in units across contests, complicating any meaningful cross-contest comparison of parameter estimates.

We restrict attention to contests reporting accuracy-based scores, which provides a common bounded score domain and facilitates a consistent transformation across contests. The retained focal paths are restricted to score observations satisfying the domain $p_t \in [0.5, 1)$ used below. To map raw leaderboard accuracies into the model’s state scale, we first transform each individual leaderboard score using the logit (log-odds) function:

$$x_t = a \cdot \log \left(\frac{p_t}{1 - p_t} \right), \quad p_t \in [0.5, 1) \quad (18)$$

where p_t is an individual leaderboard score, x_t is the corresponding transformed score, and a is a scaling parameter in the transformation.⁹ For the focal pair (i, j) , we then construct the model variables as transformed-score gaps: $\hat{y}_t = \hat{x}_t^i - \hat{x}_t^j$ for the public leaderboard and $y_t = x_t^i - x_t^j$ for the private leaderboard. We construct hourly evaluation grids by carrying forward each contestant’s most recently observed transformed score between submissions. Because submissions are asynchronous, an update by one contestant combines that contestant’s new score with the opponent’s most recently submitted score. The resulting public gap is therefore a carried-forward leaderboard gap rather than a synchronous measurement of both contestants’ contemporaneous latent outputs. Our observation equation treats this gap as an approximate readout of the latent competitive state; this approximation is most credible for focal pairs with frequent submissions and substantial overlap in active periods. On the domain used here, the log-odds transformation is increasing and convex, so as the accuracy p_t approaches 1, each marginal improvement in p_t results in a disproportionately large increase in the transformed score x_t . This feature aligns with the intuition that it becomes increasingly difficult to improve performance near the upper bound, making such gains more informative in competitive settings.

⁹ Changing a rescales the transformed score and can affect scale-dependent likelihood and validation quantities, but it does not change scale-invariant shape diagnostics of the transformed distribution. In this paper, to preserve comparability across contests and avoid extra tuning degrees of freedom, we fix $a = 1$ throughout.

5.2. Private-Leaderboard Primitive Validation

We select 73 contests from the Meta-Kaggle database, encompassing all competitions concluded before April 15, 2025 that satisfy our eligibility filters on leaderboard availability, public-private score information, and focal-player activity.¹⁰ For each selected contest, we identify two highly active top-ranked participants who meet our focal-player selection criteria. Appendix C reports the Bayesian estimation results for each contest individually.

We next conduct post-estimation diagnostic validation of the contest-specific primitives using private-leaderboard-based benchmarks.

5.2.1. Private-Leaderboard Diagnostic for Signal Precision λ . The signal precision parameter λ is contest-specific and depends on how informative public leaderboard feedback is relative to latent performance. We compare two estimates: (i) a structural estimate from submissions and public leaderboard paths, and (ii) a benchmark estimate from public-private leaderboard discrepancies. This benchmark is informative but not fully independent of the maintained structural setup, because it still relies on the transformed leaderboard scale used in the model.

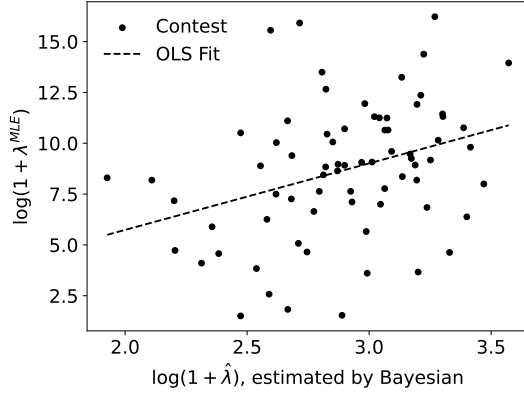
Let $\ell \in \{1, \dots, L\}$ index contests. Under the maintained Gaussian discrepancy approximation induced by the discretized signal readout, we construct the following signal-precision benchmark. Let $\mathcal{E}_\ell$ denote the ordered set of submission-triggered leaderboard readouts for the focal pair, and let $N^\ell = |\mathcal{E}_\ell|$. Let $h_k^\ell = t_k^\ell - t_{k-1}^\ell$ denote the elapsed time associated with readout t_k^ℓ . Under the maintained approximation that the discrepancy at these readouts follows $\hat{y}_{t_k}^\ell - y_{t_k}^\ell \sim \mathcal{N}(0, 1/(\lambda_\ell h_k^\ell))$, $\forall t_k \in \mathcal{E}_\ell$, we construct

$$\lambda_\ell^{\text{bench}} = \frac{N^\ell}{\sum_{t_k \in \mathcal{E}_\ell} h_k^\ell (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2}.$$

Thus, the benchmark uses only leaderboard readouts triggered by submissions, rather than repeated hourly carry-forward observations. Because contestant submissions are asynchronous, both displayed public and private gaps may combine a newly updated score for one contestant with a carried-forward score for the other. The discrepancy benchmark therefore inherits the carried-forward-gap approximation and should be interpreted as an approximate signal-precision diagnostic rather than as an exact MLE under synchronous observations.

It is important to emphasize that this benchmark relies on the Gaussian discrepancy approximation applied to the transformed scores in (18). The transformation is adopted for tractability and should be interpreted as an approximation rather than as a claim of exact normality for every contest. Auxiliary checks with alternative choices of the scale parameter a assess sensitivity to transformed-score units, not sensitivity to distributional shape. The Gaussian discrepancy structure therefore remains a maintained modeling assumption.

¹⁰ The Meta-Kaggle database is updated daily, with new competition data added after the contests conclude.



	coef.	stderr	95% CI
const.	-0.808	3.157	[-7.10, 5.49]
$\log(1 + \hat{\lambda})$	3.273**	1.081	[1.12, 5.43]
Obs.		73	
R^2		0.114	
Adj. R^2		0.102	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
(2) OLS residual diagnostics: Omnibus = 0.397, Jarque-Bera = 0.071, Skewness = 0.032, Kurtosis = 3.139. (3) Other diagnostic statistics: Condition Number = 27.9, Durbin-Watson = 1.806.

Figure 6 $\lambda_\ell^{\text{bench}}$ vs. Bayesian Estimate $\hat{\lambda}_\ell$

Note. This figure plots the approximate contest-specific signal-precision benchmark $\lambda_\ell^{\text{bench}}$ against the Bayesian structural estimate $\hat{\lambda}_\ell$ for each of the 73 contests. The benchmark is computed from public-private leaderboard discrepancies at submission-triggered readouts as $\lambda_\ell^{\text{bench}} = N^\ell / \sum_{t_k \in \mathcal{E}_\ell} h_k^\ell (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2$, where $N^\ell = |\mathcal{E}_\ell|$ is the number of readouts used in contest ℓ and $h_k^\ell = t_k^\ell - t_{k-1}^\ell$ is the elapsed time associated with readout t_k^ℓ . Because the displayed gaps are carried forward asynchronously, this quantity is motivated by the Gaussian discrepancy approximation rather than an exact synchronous-observation MLE. The dashed line represents the fitted regression line from $\log(1 + \lambda_\ell^{\text{bench}}) = \beta_0 + \beta_1 \log(1 + \hat{\lambda}_\ell) + \varepsilon_\ell$. The positive slope ($\beta_1 = 3.273$, $p < 0.01$) indicates a statistically significant but modest association ($R^2 = 0.114$).

We then regress the contest-specific private-leaderboard benchmark $\lambda_\ell^{\text{bench}}$ on the Bayesian structural estimate $\hat{\lambda}_\ell$ to examine whether a positive relationship holds between the two.

Figure 6 indicates a positive association between structural and benchmark estimates of λ . The slope on $\log(1 + \hat{\lambda})$ is positive (3.273, $p < 0.01$), but the fit is modest ($R^2 \approx 0.11$), so this should be interpreted as directional evidence rather than precise contest-level recovery. This dispersion is consistent with limited sample size and with approximation error in benchmark construction, especially the normality-sensitive transformation step. Section 6.1 provides a complementary, directional check using an alternative focal-pair definition; the positive pattern for λ also appears there.

5.2.2. Public-Private Leaderboard Disagreement Diagnostics. We next provide an additional diagnostic for signal precision based directly on public-private leaderboard disagreement, without first converting the disagreement into a Gaussian precision benchmark. If $\hat{\lambda}_\ell$ captures public-signal precision, contests with larger $\hat{\lambda}_\ell$ should exhibit smaller disagreement between the public leaderboard path and the corresponding private-leaderboard path. We therefore construct the following contest-level outcome:

$$D_\ell^{\text{path}} = \frac{1}{|\mathcal{H}_\ell|} \sum_{h \in \mathcal{H}_\ell} |y_{\ell h} - \hat{y}_{\ell h}|,$$

where $\mathcal{H}_\ell$ denotes the set of hourly observation points used in the empirical leaderboard path, and D_ℓ^{path} measures the mean absolute public-private gap over the contest path. The predicted association is therefore negative.

Because the full-path diagnostic uses the same contest path both to estimate $\hat{\lambda}_\ell$ and to construct D_ℓ^{path} , it may partly reflect common path-level variation rather than purely predictive content. We therefore also report a time-split version of this diagnostic. Let $\hat{\lambda}_\ell^{early}$ denote the structural estimate obtained using only the first τT_ℓ hours of contest-phase data, with $\tau = 0.75$. The validation target is then constructed only from the remaining public-private path:

$$D_\ell^{late} = \frac{1}{|\mathcal{H}_\ell^{late}|} \sum_{h \in \mathcal{H}_\ell^{late}} |y_{\ell h} - \hat{y}_{\ell h}|.$$

Here $\mathcal{H}_\ell^{late} = \{h \in \mathcal{H}_\ell : h > \tau T_\ell\}$. For both diagnostics, we estimate the log-log validation slope linking the discrepancy measure to the corresponding signal-precision estimate. The full-path specification uses $\log(1 + D_\ell^{path})$ and $\log(1 + \hat{\lambda}_\ell)$, while the time-split specification uses $\log(1 + D_\ell^{late})$ and $\log(1 + \hat{\lambda}_\ell^{early})$. The latter asks whether signal precision estimated from early public contest information predicts public-private disagreement measured over the later contest period, conditional on the focal-pair selection rule.

Table 5 Signal Precision and Public-Private Path Disagreement

Specification	Coef.	95% CI	R^2	Spearman ρ
Full path	-0.156* (0.063)	[-0.282, -0.030]	0.079	-0.340**
Time split	-0.046 (0.092)	[-0.225, 0.134]	0.003	-0.220
Obs.	73			

Note: For contest ℓ , $D_\ell^{path} = |\mathcal{H}_\ell|^{-1} \sum_{h \in \mathcal{H}_\ell} |y_{\ell h} - \hat{y}_{\ell h}|$ measures the mean absolute gap between private and public leaderboard paths. The full-path row reports the slope from $\log(1 + D_\ell^{path}) = \alpha + \beta \log(1 + \hat{\lambda}_\ell) + \varepsilon_\ell$, where $\hat{\lambda}_\ell$ is the baseline structural estimate. For the time-split row, $\hat{\lambda}_\ell^{early}$ is estimated using only the first 75% of contest-phase data, and $D_\ell^{late} = |\mathcal{H}_\ell^{late}|^{-1} \sum_{h \in \mathcal{H}_\ell^{late}} |y_{\ell h} - \hat{y}_{\ell h}|$ is measured using only the remaining 25% of the private and public leaderboard paths. The reported coefficient is the slope from $\log(1 + D_\ell^{late}) = \alpha + \beta \log(1 + \hat{\lambda}_\ell^{early}) + \varepsilon_\ell$. Standard errors are reported in parentheses under the coefficient estimates. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 5 reports both specifications. The full-path diagnostic has the predicted negative slope (-0.156 , $p < 0.05$) and a negative Spearman correlation (-0.340 , $p < 0.01$), consistent with the view that higher estimated signal precision is associated with smaller public-private leaderboard disagreement. The time-split estimate has the same sign but is smaller and imprecise, so it should be interpreted as directional rather than confirmatory evidence. Taken together with the benchmark-precision exercise above, these results provide suggestive diagnostic evidence for the λ interpretation: the predicted associations appear in the full-sample specifications, but signal precision is not precisely recovered at the contest level. The weaker time-split result is expected because $\hat{\lambda}_\ell^{early}$ is

estimated from less data and D_ℓ^{late} is measured over a shorter validation window, both of which increase sampling variability.

5.2.3. Private-Leaderboard Consistency Benchmark for Innovation Risk σ . Another key contest-specific parameter is σ , which captures innovation uncertainty and the extent to which contest outcomes are driven by chance rather than effort. For each contest ℓ , we estimate $\hat{\sigma}_\ell$ structurally using the two contestants' public submission records $\{t_k^\ell\}_{k=1}^{N_i^\ell+N_j^\ell}$ and public leaderboard trajectories $\{\hat{y}_k^\ell\}_{k=1}^{N_i^\ell+N_j^\ell}$. We then construct a private-leaderboard-based benchmark for σ using score increments.

Let $(t_k^\ell, y_k^\ell)_{k=0}^{n_\ell}$ denote the private-leaderboard gap evaluated at the ordered update times used for this benchmark, including the normalized initial observation. Under the diffusion specification in (1), the private-leaderboard increment satisfies $y_k^\ell - y_{k-1}^\ell \mid \{m_i^\ell, m_j^\ell\} \sim \mathcal{N}(\mu_k^\ell, \sigma^2(t_k^\ell - t_{k-1}^\ell))$ for $k = 1, \dots, n_\ell$. Here, $\mu_k^\ell = \int_{t_{k-1}^\ell}^{t_k^\ell} [m_i^\ell(s) - m_j^\ell(s)] ds$. Motivated by the Gaussian increment formula, we compute the following contest-specific consistency benchmark; the corresponding diffusion calculation is provided in Appendix B.2:

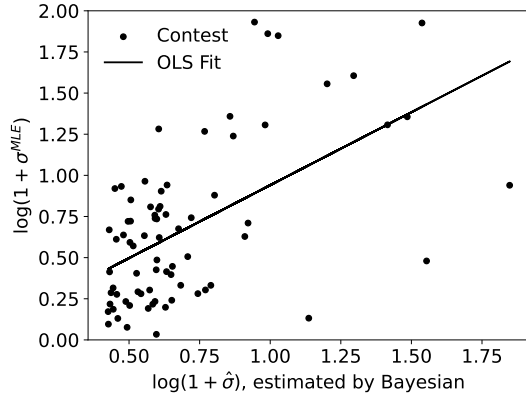
$$\sigma_\ell^{\text{bench}} = \left(\frac{1}{n_\ell} \sum_{k=1}^{n_\ell} \frac{[y_k^\ell - y_{k-1}^\ell - \hat{\mu}_k^\ell]^2}{t_k^\ell - t_{k-1}^\ell} \right)^{1/2}$$

The drift adjustment $\hat{\mu}_k^\ell$ is computed from the Bayesian posterior effort trajectories, so this benchmark is model-assisted rather than independent. In addition, because contestant submissions are asynchronous, consecutive displayed private-gap observations may combine a new score for one contestant with a carried-forward score for the other. The resulting estimator should therefore be interpreted as an approximate consistency benchmark, not as an exact maximum likelihood estimator of latent diffusion volatility. For this reason, we place greater weight on the less model-dependent private-path volatility diagnostics reported below.

As in the analysis above for λ , we regress the benchmark σ^{bench} on the structural estimate $\hat{\sigma}$ to examine whether a positive relationship exists between the two.

Figure 7 reveals a positive association between the structural estimate and the private-leaderboard-based volatility benchmark. The intercept is small and insignificant, while the slope on $\log(1 + \hat{\sigma})$ is 0.887 ($p < 0.001$), with $R^2 \approx 0.317$. Thus, the estimated innovation-risk primitive is positively associated with an approximate private-leaderboard consistency benchmark whose path values are excluded from the likelihood.

5.2.4. Private-Leaderboard Volatility Diagnostics. The preceding validation of σ uses an approximate private-leaderboard consistency benchmark motivated by the normal increment structure of the diffusion model. That benchmark is parametric and uses the posterior drift adjustment



	coef.	stderr	95% CI
const.	0.054	0.116	[-0.18, 0.29]
$\log(1 + \hat{\sigma})$	0.887***	0.152	[0.58, 1.19]
Obs.	73		
R^2	0.317		
Adj. R^2	0.308		

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
(2) OLS residual diagnostics: Omnibus = 1.280, Jarque-Bera = 0.774, Skewness = 0.225, Kurtosis = 3.214. (3) Other diagnostic statistics: Condition Number = 5.08, Durbin-Watson = 1.618.

Figure 7 σ^{bench} vs. Bayesian Estimate $\hat{\sigma}$

Note. This figure plots the private-leaderboard consistency benchmark σ^{bench} against the Bayesian structural estimate $\hat{\sigma}$ for each of the 73 contests. The benchmark is computed from carried-forward private leaderboard increments as $\sigma^{\text{bench}} = \left(\frac{1}{n} \sum_{k=1}^n \frac{(y_k - y_{k-1} - \hat{\mu}_k)^2}{t_k - t_{k-1}} \right)^{1/2}$, where n is the number of adjacent displayed-gap increments and $\hat{\mu}_k$ is computed from posterior effort trajectories. Because asynchronous observations can combine a new score with an opponent's stale score, the quantity is an approximate, model-assisted consistency benchmark rather than an exact diffusion MLE. The dashed line represents the fitted regression line from $\log(1 + \sigma^{\text{bench}}) = \beta_0 + \beta_1 \log(1 + \hat{\sigma}) + \varepsilon$.

$\hat{\mu}_k$. We therefore add a simpler diagnostic based on realized private-leaderboard volatility. For each contest ℓ , let $\Delta y_{\ell h} = y_{\ell h} - y_{\ell, h-1}$ denote the increment on the constructed hourly evaluation grid, and define $V_{\ell}^{\text{path}} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_{\ell}, h > 1\}$. This statistic does not require estimating a normal-increment likelihood or subtracting a model-implied drift term. It asks whether contests with larger estimated innovation risk also exhibit more volatile private-leaderboard paths. Because the hourly private paths are constructed by carrying scores forward between submissions, the resulting series contains zero increments between updates and discrete jumps at submission times. This volatility measure therefore reflects both changes in submitted scores and the timing of private-leaderboard updates. We interpret it as a less model-dependent path diagnostic, not as a direct measure of latent diffusion volatility. We estimate this relationship in logs, using $\log(1 + V_{\ell}^{\text{path}})$ and $\log(1 + \hat{\sigma}_{\ell})$, with the predicted positive coefficient.

We also report a time-split version of the same diagnostic. Let $\hat{\sigma}_{\ell}^{\text{early}}$ denote the structural estimate obtained using only the first τT_{ℓ} hours of contest-phase data, with $\tau = 0.75$. The validation target is then constructed only from the remaining private-leaderboard path, $V_{\ell}^{\text{late}} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_{\ell}, h > \tau T_{\ell}\}$. For the time-split specification, we use $\log(1 + V_{\ell}^{\text{late}})$ and $\log(1 + \hat{\sigma}_{\ell}^{\text{early}})$. Conditional on the focal-pair selection rule, the time-split version provides a stricter temporal separation because the structural estimate uses only early public contest data, while the validation outcome is measured on the later private-leaderboard path.

Table 6 shows a positive relationship between estimated innovation risk and realized private-leaderboard volatility. The full-path slope is 0.278 ($p < 0.001$), with $R^2 = 0.356$ and a Spearman

Table 6 Private-Leaderboard Volatility Diagnostics

Specification	Coef.	95% CI	R^2	Spearman ρ
Full path	0.278*** (0.078)	[0.126, 0.430]	0.356	0.492***
Time split	0.220* (0.101)	[0.022, 0.418]	0.153	0.314**
Obs.	73			

Note: For contest ℓ , $y_{\ell h}$ denotes the private-leaderboard gap on the constructed hourly evaluation grid, and $\Delta y_{\ell h} = y_{\ell h} - y_{\ell, h-1}$ is the corresponding grid increment. The full-path row reports the slope β from $\log(1 + V_{\ell}^{path}) = \alpha + \beta \log(1 + \hat{\sigma}_{\ell}) + \varepsilon_{\ell}$, where $V_{\ell}^{path} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_{\ell}, h > 1\}$ and $\hat{\sigma}_{\ell}$ is the baseline structural estimate. The time-split row reports the slope β from $\log(1 + V_{\ell}^{late}) = \alpha + \beta \log(1 + \hat{\sigma}_{\ell}^{early}) + \varepsilon_{\ell}$, where $\hat{\sigma}_{\ell}^{early}$ is estimated using only the first 75% of contest-phase data and $V_{\ell}^{late} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_{\ell}, h > 0.75T_{\ell}\}$. Spearman ρ is the rank correlation between the same dependent variable and structural estimate used in each row. Standard errors are reported in parentheses under the coefficient estimates. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

correlation of 0.492 ($p < 0.001$). The time-split slope is smaller but remains positive and significant: 0.220 ($p < 0.05$), with a Spearman correlation of 0.314 ($p < 0.01$). These results complement the normal-increment consistency benchmark and suggest that the σ estimates capture meaningful variation in private-leaderboard volatility. Because both $\hat{\sigma}_{\ell}$ and the private-path volatility measures are expressed in transformed-score units, their cross-contest association may also reflect contest-level scale variation. We therefore interpret these regressions as diagnostics rather than scale-free validation evidence.

5.3. Counterfactual Prize-Duration Substitution

The validation exercises suggest that the estimated primitives contain meaningful information about the contest environment. We use these primitives to revisit the design question in Section 2.1. Under the maintained approximation, prize and duration are the two design variables that directly affect predicted total effort.¹¹ Because the designer's cardinal payoff and the platform cost of longer contests are difficult to calibrate, we do not solve for a welfare-optimal design. Instead, we trace the model-implied substitution between prize spending and contest duration while holding predicted total effort fixed.

For each contest, let (θ^{obs}, T^{obs}) denote the observed prize and duration, and let $M_0 = M(\theta^{obs}, T^{obs})$ denote predicted total effort under the observed design. For a duration multiplier $a \geq 1$, let $T_a = aT^{obs}$ and define

$$\theta_a = \min \{ \theta \in [\underline{\theta}, \theta^{obs}] : M(\theta, T_a) \geq M_0 \},$$

¹¹ We do not emphasize a baseline information-design counterfactual for λ . As shown in Section 2.1, under the maintained steady-state filtering restriction $S_t \equiv \bar{S}$, signal precision does not generate an independent comparative-static effect on total expected effort. Studying optimal information disclosure would require relaxing the steady-state restriction and allowing filtering uncertainty to evolve endogenously over time.

where $\underline{\theta} = 0.01$ thousand USD is the lower prize bound used in the numerical search. Under the maintained monotonicity of predicted effort in duration, the feasible set is nonempty because $\theta = \theta^{obs}$ satisfies the effort constraint for $a \geq 1$. The required prize change is $100(\theta_a - \theta^{obs})/\theta^{obs}$, so negative values indicate prize reductions that keep predicted total effort at least at the observed-design baseline. This exercise is not a claim that longer contests are costless or welfare improving. Rather, it reports the iso-effort prize-duration frontier: how much prize spending can be substituted by a small, externally specified increase in duration.

Table 7 reports the substitution frontier for duration multipliers close to the observed design. When $a = 1$, the counterfactual coincides with the observed design and no prize reduction is possible. For each positive duration increase considered in Table 7, predicted total effort can be preserved with a lower prize in all 73 contests. This direction is a direct implication of monotonicity of predicted total effort in both prize and duration; the empirical content is the magnitude of the required prize adjustment. A one percent increase in duration allows a median prize reduction of 1.23%, while a ten percent increase allows a median prize reduction of 11.03%. The corresponding means are substantially larger than the medians, indicating a strongly right-skewed distribution of substitution rates. We therefore emphasize the medians as the more representative summary.

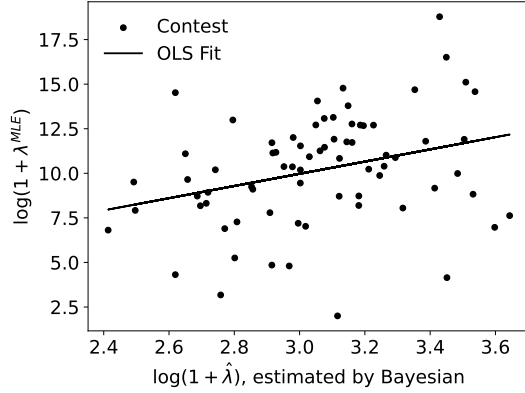
Table 7 Counterfactual Prize-Duration Substitution

Duration multiplier	Lower prize	$\geq 1\%$ lower	$\geq 5\%$ lower	$\geq 10\%$ lower	Mean change	Median change
1.00	0/73	0/73	0/73	0/73	0.00%	0.00%
1.01	73/73	42/73	25/73	21/73	-11.09%	-1.23%
1.02	73/73	69/73	32/73	24/73	-12.50%	-2.44%
1.05	73/73	73/73	41/73	31/73	-15.86%	-5.86%
1.10	73/73	73/73	68/73	40/73	-20.02%	-11.03%

Note: This table reports counterfactual prize-duration substitution results for the 73 contests with complete observed design inputs and structural estimates. For duration multiplier a , the counterfactual duration is $T_a = aT^{obs}$. For each contest, we compute the minimum prize $\theta_a \in [0.01, \theta^{obs}]$ thousand USD such that predicted total effort remains weakly above $M(\theta^{obs}, T^{obs})$. “Lower prize” counts contests with $\theta_a < \theta^{obs}$. The next three columns count contests in which the required prize is at least 1%, 5%, or 10% below the observed prize. Mean and median changes report $100(\theta_a - \theta^{obs})/\theta^{obs}$, so negative values denote required prize reductions.

6. Robustness

This section examines the sensitivity of our main validation results to key modeling choices. We focus on two concerns highlighted in Section 3.2: the selection of the two-player focal pair and the normality assumption underlying the leaderboard transformation. For each robustness check, we re-estimate the model and compare the resulting validation patterns to the baseline findings reported in Section 5.2.



	coef.	stderr	95% CI
const.	-0.290	3.819	[-7.91, 7.32]
$\log(1 + \hat{\lambda})$	3.421**	1.244	[0.94, 5.90]
Obs.		73	
R^2		0.096	
Adj. R^2		0.084	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
(2) OLS residual diagnostics: Omnibus = 3.192, Jarque-Bera = 2.424, Skewness = -0.411, Kurtosis = 3.350. (3) Other diagnostic statistics: Condition Number = 36.9, Durbin-Watson = 2.236.

Figure 8 Robustness Check (2nd–3rd Pair): $\lambda_\ell^{\text{bench}}$ vs. Bayesian Estimate $\hat{\lambda}_\ell$

Note. This figure reports the validation comparison under an alternative focal pair: the second- and third-ranked participants (based on final private leaderboard ranks) instead of the top two. The dashed line is the fitted regression $\log(1 + \lambda_\ell^{\text{bench}}) = \beta_0 + \beta_1 \log(1 + \hat{\lambda}_\ell) + \varepsilon_\ell$, with $\beta_1 = 3.421$ ($p < 0.01$) and $R^2 = 0.096$. The positive slope remains significant, though the fit is modest, so the result is best interpreted as directional evidence under an alternative focal-pair definition.

6.1. Selection of Two Contestants

The baseline empirical analysis pairs two highly active top-ranked contestants in each contest. To assess whether the validation results are sensitive to this focal-pair choice, we re-estimate the model under an alternative pairing rule.

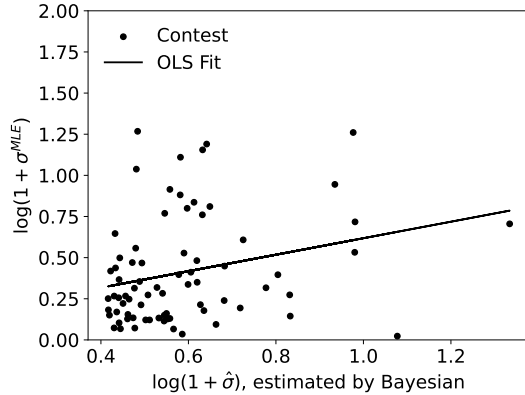
As a robustness check, we estimate the model on the second- and third-ranked participants in each contest, based on final private leaderboard ranks. This specification shifts the focal rivalry away from the winner and tests whether the validation pattern is driven by one specific top-pair definition.

For signal precision, the alternative focal-pair specification based on the second and third ranked participants again yields a positive and significant relationship, although the explanatory power remains limited ($R^2 = 0.096$). This check asks whether changing the focal-pair definition overturns the qualitative validation pattern. It does not: the sign and significance are preserved, while the low fit reinforces the more cautious interpretation that signal precision is harder to recover precisely at the contest level.

For innovation risk, the positive and significant association also survives this alternative pairing rule, although the fit is weaker than in the baseline specification. Together, these results suggest that the signs and significance patterns of the main validation exercises are not driven solely by one focal-pair definition. At the same time, the strength of the association, especially for σ , is sensitive to the selected pair.

6.2. Normality Caveat and Interpretation

The multiplicative parameter a in equation (18) rescales transformed scores and can affect scale-dependent likelihood and validation quantities, but it cannot alter scale-invariant normality diag-



	coef.	stderr	95% CI
const.	0.117	0.130	[-0.14, 0.38]
$\log(1 + \hat{\sigma})$	0.501*	0.212	[0.08, 0.92]
Obs.	73		
R^2	0.071		
Adj. R^2	0.059		

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

(2) OLS residual diagnostics: Omnibus = 12.72, Jarque-Bera = 13.66, Skewness = 1.02, Kurtosis = 3.50. (3) Other diagnostic statistics: Condition Number = 7.84, Durbin-Watson = 1.87.

Figure 9 Robustness Check (2nd–3rd Pair): σ^{bench} vs. Bayesian Estimate $\hat{\sigma}$

Note. This figure replicates the validation exercise in Figure 7 using an alternative focal pair: the second- and third-ranked participants (based on final private leaderboard ranks) instead of the top two. The dashed line is the fitted regression $\log(1 + \sigma^{\text{bench}}) = \beta_0 + \beta_1 \log(1 + \hat{\sigma}) + \varepsilon$, with $\beta_1 = 0.501$ ($p < 0.05$) and $R^2 = 0.071$. The positive slope remains significant, though the association is weaker than in the baseline ($\beta_1 = 0.887$, $R^2 = 0.317$), indicating that σ recovery is directionally robust but sensitive to the choice of focal pair.

nostics. Auxiliary checks across alternative values of a should therefore be interpreted as scale-sensitivity checks, not as evidence for the Gaussian approximation. Additional scale-sensitivity results are available from the authors.

We do not conduct a fully non-Gaussian robustness analysis here, because both the benchmark constructions and the Bayesian estimation procedure rely on the maintained Gaussian structure. Extending the analysis beyond this setting would require re-specifying the likelihood and re-deriving the benchmark estimators under alternative distributional assumptions.

Overall, the robustness analyses support a disciplined interpretation: the two-player reduction is a useful empirical summary of focal rivalry rather than a literal description of full contest dynamics, and the private-leaderboard validation results should be interpreted as directional evidence under the maintained transformation and Gaussian approximation.

7. Conclusion

This paper develops a dynamic game-theoretic model and a Bayesian structural estimation framework for online crowdsourcing contests under noisy public feedback. Conditional on focal pairs selected using final private ranks, we recover latent strategic objects using only contest-phase submission timing and public leaderboard trajectories. We then assess the estimated contest-specific primitives with private-leaderboard path benchmarks excluded from the likelihood. Across 73 Kaggle contests, these path-wise holdout diagnostics provide disciplined evidence on the estimated primitives. The evidence is strongest for innovation risk, where both benchmark and time-split volatility diagnostics support the structural estimates; for signal precision, the evidence is directionally consistent but more modest. In complementary synthetic exercises, the initial-condition

parameter μ_0 shows little posterior concentration relative to the main contest-level primitives. The counterfactual exercise traces the model-implied prize-duration substitution frontier, rather than providing a full welfare comparison of observed and alternative contest designs.

Nonetheless, several limitations deserve mention. First, the estimation relies on assumptions about the data-generating process (DGP) that may not fully reflect real-world complexities. For example, (1) many contests use multi-tier or team-based prizes rather than the winner-take-all format assumed here; (2) we focus on two-player settings, while actual contests involve broader participation; (3) leaderboard scores vary in format, complicating their integration into a unified signal structure; and (4) the focal pair is selected using final private ranks, so the private-path validation is conditional on ex post pair selection rather than fully independent. Second, the empirical filtering implementation involves two additional approximations. The estimation method simplifies belief updating by assuming the inference variance quickly converges and remains constant. A more realistic approach would allow inference precision to evolve over time. The empirical implementation also approximates asynchronous, submission-triggered leaderboard updates with a continuous-update filtering equation. A fully coherent hybrid filter with discrete measurement updates would better represent stale scores and irregular observation times but would sacrifice much of the tractability of the current framework. Third, cross-contest comparisons of innovation risk and private-path volatility may retain a shared transformed-score scale component despite the common logit mapping. These limitations point to valuable directions for future work, including richer prize structures, multi-agent dynamics, and more flexible learning mechanisms.

Appendix A: Proofs

Proof of Proposition 1 Under the symmetric benchmark $c_i = c_j = c$, equation (9) implies that the leading-order approximation to total expected effort at $t = 0$ is

$$\bar{M}(\tilde{y}_0, 0) = \frac{2\theta\sqrt{T}}{\sigma c\sqrt{2\pi}} \exp\left(-\frac{\tilde{y}_0^2}{2\sigma^2 T}\right).$$

For notational convenience, define $A = 2/(\sigma c\sqrt{2\pi})$ and $B = \tilde{y}_0^2/(2\sigma^2)$. Then

$$\bar{M}(\tilde{y}_0, 0) = A\theta T^{1/2} \exp(-B/T).$$

Since λ does not enter $\bar{M}(\tilde{y}_0, 0)$, the approximated total effort is neutral with respect to the signal precision λ . Moreover, because the designer's objective is given by $\Pi(\theta, T, \lambda) = \beta^T \bar{M}(\tilde{y}_0, 0)^\alpha - \theta$, it is also neutral with respect to λ . It remains to establish the dependence on θ and T .

We first prove the monotonicity of $\bar{M}(\tilde{y}_0, 0)$. Differentiating with respect to θ gives $\partial \bar{M}(\tilde{y}_0, 0)/\partial \theta = AT^{1/2} \exp(-B/T) > 0$, because $A > 0$ and $T > 0$. Hence, $\bar{M}(\tilde{y}_0, 0)$ is increasing in the prize amount θ . Next, differentiating with respect to T gives

$$\frac{\partial \bar{M}(\tilde{y}_0, 0)}{\partial T} = A\theta \exp(-B/T) \left(\frac{1}{2} T^{-1/2} + BT^{-3/2} \right).$$

Since $A > 0$, $\theta > 0$, $B \geq 0$, and $T > 0$, we have $\partial \bar{M}(\tilde{y}_0, 0)/\partial T > 0$. Therefore, $\bar{M}(\tilde{y}_0, 0)$ is increasing in the contest duration T .

We now turn to the contest designer's objective. Using the expression above, we can write

$$\Pi(\theta, T, \lambda) = A^\alpha \theta^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T) - \theta.$$

First, fix $T > 0$. Define $D(T) = AT^{1/2} \exp(-B/T)$. Then $\bar{M}(\tilde{y}_0, 0) = D(T)\theta$, and the objective becomes

$$\Pi(\theta, T, \lambda) = \beta^T D(T)^\alpha \theta^\alpha - \theta.$$

Differentiating with respect to θ gives $\partial \Pi(\theta, T, \lambda)/\partial \theta = \alpha \beta^T D(T)^\alpha \theta^{\alpha-1} - 1$. The second derivative is

$$\frac{\partial^2 \Pi(\theta, T, \lambda)}{\partial \theta^2} = \alpha(\alpha - 1) \beta^T D(T)^\alpha \theta^{\alpha-2} < 0,$$

because $0 < \alpha < 1$. Hence, for any fixed $T > 0$, the objective is strictly concave in θ . Moreover, as $\theta \downarrow 0$, we have $\theta^{\alpha-1} \rightarrow +\infty$, so $\partial \Pi/\partial \theta \rightarrow +\infty$; as $\theta \rightarrow +\infty$, we have $\theta^{\alpha-1} \rightarrow 0$, so $\partial \Pi/\partial \theta \rightarrow -1$. Therefore, there exists a unique maximizer in θ , and the objective is single-peaked in the prize amount θ .

Next, fix $\theta > 0$ and consider the dependence on T . Up to the constant term $-\theta$, the objective can be written as

$$\Pi(\theta, T, \lambda) + \theta = (A\theta)^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T).$$

Let $G(T) = \log(\Pi(\theta, T, \lambda) + \theta)$. Since $\Pi(\theta, T, \lambda) + \theta > 0$, the sign of $\partial \Pi/\partial T$ is the same as the sign of $G'(T)$. We have

$$G(T) = \alpha \log(A\theta) + T \log \beta + (\alpha/2) \log T - \alpha B/T.$$

Therefore, $G'(T) = \log \beta + \alpha/(2T) + \alpha B/T^2$.

If $\beta = 1$, then $\log \beta = 0$, and hence $G'(T) = \alpha/(2T) + \alpha B/T^2 > 0$. Thus, when $\beta = 1$, the objective is increasing in T .

If $0 < \beta < 1$, then $\log \beta < 0$. Moreover,

$$G''(T) = -\alpha/(2T^2) - 2\alpha B/T^3 < 0.$$

Thus, $G'(T)$ is strictly decreasing in T . As $T \downarrow 0$, $G'(T) \rightarrow +\infty$, while as $T \rightarrow +\infty$, $G'(T) \rightarrow \log \beta < 0$. Hence, there exists a unique $T^* > 0$ such that $G'(T^*) = 0$. Since the sign of $\partial \Pi/\partial T$ is the same as the sign of $G'(T)$, the objective increases for $T < T^*$ and decreases for $T > T^*$. Therefore, if $0 < \beta < 1$, the objective is single-peaked in the contest duration T .

Finally, consider the joint optimization over (θ, T) . For each fixed $T > 0$, the first-order condition with respect to θ is

$$\alpha A^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T) \theta^{\alpha-1} = 1.$$

Because $0 < \alpha < 1$, this condition has a unique solution, denoted by $\theta^*(T)$, given by

$$\theta^*(T) = [\alpha A^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T)]^{1/(1-\alpha)}.$$

The strict concavity in θ established above implies that $\theta^*(T)$ is the unique optimal prize conditional on T .

We can therefore study the profile objective $\Pi(\theta^*(T), T, \lambda)$. By the first-order condition, $A^\alpha \beta^T T^{\alpha/2} \exp(-\alpha B/T) (\theta^*(T))^\alpha = \theta^*(T)/\alpha$. Hence,

$$\Pi(\theta^*(T), T, \lambda) = \frac{1-\alpha}{\alpha} \theta^*(T).$$

Since $(1-\alpha)/\alpha > 0$, maximizing the profile objective over T is equivalent to maximizing $\theta^*(T)$. Because the exponent $1/(1-\alpha)$ is positive, this is equivalent to maximizing $\beta^T T^{\alpha/2} \exp(-\alpha B/T)$ over $T > 0$.

Let $H(T) = T \log \beta + (\alpha/2) \log T - \alpha B/T$. Then maximizing $\beta^T T^{\alpha/2} \exp(-\alpha B/T)$ is equivalent to maximizing $H(T)$. Differentiating gives $H'(T) = \log \beta + \alpha/(2T) + \alpha B/T^2$, and

$$H''(T) = -\alpha/(2T^2) - 2\alpha B/T^3 < 0.$$

Thus, $H(T)$ is strictly concave on $T > 0$.

If $0 < \beta < 1$, then $\log \beta < 0$. Moreover, as $T \downarrow 0$, $H'(T) \rightarrow +\infty$, while as $T \rightarrow +\infty$, $H'(T) \rightarrow \log \beta < 0$. Therefore, there exists a unique $T^* > 0$ such that $H'(T^*) = 0$. The corresponding optimal prize is $\theta^* = \theta^*(T^*)$. Hence, when $0 < \beta < 1$, the joint problem admits a unique interior maximizer (θ^*, T^*) .

If $\beta = 1$, then $H'(T) = \alpha/(2T) + \alpha B/T^2 > 0$ for all $T > 0$. Therefore, the profile objective is increasing in T , and no finite interior maximizer in T exists unless an upper bound on contest duration is imposed. If an upper bound $T \leq \bar{T}$ is imposed, the optimal duration is $T^* = \bar{T}$, and the optimal prize is $\theta^*(\bar{T})$. $\square$

Proof of Lemma 1 The proof proceeds under the maintained structural specification in (6), (14), (15), and (16). Suppose two parameter vectors $\Theta = (c_i, c_j, \mu_0, \sigma, \lambda)$ and $\Theta' = (c'_i, c'_j, \mu'_0, \sigma', \lambda')$ induce the same likelihood in a neighborhood of the true parameter value.

First, consider the Gaussian leaderboard component in (16). Equality of Gaussian likelihoods for all sampled paths implies equality of both covariance and mean structures. Hence,

$$\Sigma_y + D_\lambda = \Sigma'_y + D_{\lambda'}, \quad \mu_k = \mu'_k \text{ for all } k.$$

Here D_λ and $D_{\lambda'}$ are diagonal matrices with $D_{\lambda, kk} = 1/(\lambda h_k)$ and $D_{\lambda', kk} = 1/(\lambda' h_k)$, where the readout intervals $\{h_k\}_{k=1}^N$ are fixed by the observation design. With at least two distinct leaderboard observations, the off-diagonal covariance terms equal $\sigma^2 \min(t_i, t_j)$ and are unaffected by the independent signal-noise term. Therefore equality of the off-diagonal covariance terms implies $\sigma = \sigma'$. Given $\sigma = \sigma'$, equality of the diagonal terms then implies $\lambda = \lambda'$.

It remains to identify (c_i, c_j, μ_0) . Given (σ, λ) and fixed r , define $\vartheta = (c_i, c_j, \mu_0)$. For any candidate ϑ , the filtering recursion and equilibrium effort rule generate a perceived-state path $\tilde{y}_t(\vartheta)$, effort paths $m_i(t; \vartheta)$ and $m_j(t; \vartheta)$, submission intensities

$$\tau_i(t; \vartheta) = r m_i(t; \vartheta), \quad \tau_j(t; \vartheta) = r m_j(t; \vartheta),$$

and public-gap mean values $\mu_k(\vartheta)$ at the observed leaderboard update times. Let

$$G(\vartheta) = (\{\tau_i(t; \vartheta)\}_{t \in \mathcal{T}_i}, \{\tau_j(t; \vartheta)\}_{t \in \mathcal{T}_j}, \{\mu_k(\vartheta)\}_{k=1}^N)$$

denote this reduced-form prediction map, evaluated on the support of observed submission and update times.

We impose the full-rank variation condition that the Jacobian $\partial G(\vartheta_0)/\partial \vartheta$ has full column rank at the true parameter value ϑ_0 . This condition formalizes the requirement that the realized public-history support contain enough variation to distinguish changes in effort costs from changes in the initial perceived gap. Under this condition, equality of the point-process intensity functions and equality of the Gaussian mean structure imply $G(\vartheta) = G(\vartheta_0)$ only locally at $\vartheta = \vartheta_0$. Hence (c_i, c_j, μ_0) is locally identified.

Combining this local identification of (c_i, c_j, μ_0) with the covariance-based identification of (σ, λ) yields local identification of $(c_i, c_j, \sigma, \lambda, \mu_0)$ conditional on fixed r . $\square$

Appendix B: Auxiliary Results

B.1. Solve S_t in Equation (4)

If S_t is in steady state, $dS_t/dt = 0 \Leftrightarrow S_t = \bar{S} \equiv \sigma/\sqrt{\lambda}$. If S_t is not in steady state, i.e., $S_t \neq \bar{S}$, we first isolate the two variables and get

$$\frac{dS}{\sigma^2 - \lambda S^2} = dt$$

Integrating both sides gives

$$t = \int \frac{dS}{\sigma^2 - \lambda S^2} = \frac{1}{\sigma\sqrt{\lambda}} \int \frac{d(S\sqrt{\lambda}/\sigma)}{1 - (S\sqrt{\lambda}/\sigma)^2} \equiv \frac{1}{\sigma\sqrt{\lambda}} \int \frac{du}{1 - u^2},$$

where $u = S\sqrt{\lambda}/\sigma = S/\bar{S}$. For $\tanh^{-1}(u)$ we require $|u| < 1$ ($S < \bar{S}$), and for $\coth^{-1}(u)$ we require $|u| > 1$ ($S > \bar{S}$). Hence,

$$\sigma\sqrt{\lambda} \cdot t = \begin{cases} \tanh^{-1}(u) - K_1, & \text{if } |u| < 1 \\ \coth^{-1}(u) - K_2, & \text{if } |u| > 1 \end{cases} = \begin{cases} \tanh^{-1}(S/\bar{S}) - K_1, & \text{if } S < \bar{S} \\ \coth^{-1}(S/\bar{S}) - K_2, & \text{if } S > \bar{S} \end{cases}$$

Thus, in the non-steady-state case,

$$S = \begin{cases} \bar{S} \cdot \tanh(\sigma\sqrt{\lambda} \cdot t + K_1), & \text{if } S < \bar{S} \\ \bar{S} \cdot \coth(\sigma\sqrt{\lambda} \cdot t + K_2), & \text{if } S > \bar{S} \end{cases}$$

Finally, we determine the constants K_1, K_2 by the initial condition S_0 and have

$$K_1 = \tanh^{-1}(S_0/\bar{S}), \quad K_2 = \coth^{-1}(S_0/\bar{S})$$

If $\lambda = 0$, we have $S_t = S_0 + \sigma^2 t$, i.e., the estimation variance increases linearly over time. If $\lambda > 0$, the solution of (4) is

$$S_t = \begin{cases} \bar{S} \cdot \tanh \left\{ t \cdot \sigma\sqrt{\lambda} + \tanh^{-1}(S_0/\bar{S}) \right\} & \text{if } S_0 < \bar{S} \\ \bar{S} & \text{if } S_0 = \bar{S} \\ \bar{S} \cdot \coth \left\{ t \cdot \sigma\sqrt{\lambda} + \coth^{-1}(S_0/\bar{S}) \right\} & \text{if } S_0 > \bar{S} \end{cases}$$

Specifically, $\bar{S} = \sigma/\sqrt{\lambda}$ when $\lambda > 0$ and $\bar{S} = \infty$ when $\lambda = 0$.

B.2. Consistency Benchmark for σ^2

This appendix derives the private-leaderboard benchmark for σ used in Section 5.2. For a given contest, let $\Delta y_k = y_k - y_{k-1}$ denote the private-leaderboard gap increment between two adjacent update times, and let $h_k = t_k - t_{k-1}$ be the corresponding elapsed time. The posterior effort trajectories provide a plug-in drift adjustment

$$\hat{\mu}_k = \int_{t_{k-1}}^{t_k} [\hat{m}_i(s) - \hat{m}_j(s)] ds.$$

Under the diffusion approximation, the drift-adjusted increment satisfies

$$\Delta y_k - \hat{\mu}_k \sim \mathcal{N}(0, \sigma^2 h_k), \quad k = 1, \dots, n.$$

Treating these increments as approximately independent, the log-likelihood for σ^2 is, up to constants,

$$\ell(\sigma^2) = -\frac{n}{2} \log(\sigma^2) - \frac{1}{2} \sum_{k=1}^n \log h_k - \frac{1}{2\sigma^2} \sum_{k=1}^n \frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}.$$

Differentiating with respect to σ^2 and setting the derivative equal to zero gives

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{k=1}^n \frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k} = 0.$$

Therefore,

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^n \frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}. \quad (19)$$

In the empirical application, $\hat{\mu}_k$ is computed from the Bayesian posterior effort paths, so this benchmark is a plug-in private-leaderboard consistency check rather than a fully model-free estimator. Moreover, asynchronous update timing means that adjacent displayed gaps may combine a new score for one contestant with a carried-forward score for the other. The expression in (19) is therefore an approximate benchmark for the carried-forward gap process, not an exact maximum likelihood estimator for synchronously observed latent diffusion increments.

Appendix C: Statistics

Table 8 reports raw contest-level estimates for transparency, even though some benchmark values are highly dispersed. Large values of λ^{bench} can arise mechanically when public-private score discrepancies are very small, because the benchmark is inverse-quadratic in those discrepancies. Accordingly, cross-contest comparisons of λ^{bench} are most interpretable on a transformed scale (as used in the main-text regressions).

Table 8: Contest-Level Structural Estimates and Benchmarks

Contest Id	N_i	N_j	Prize (k, USD)	Public Data %	Bayesian Posterior Mean						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
2445	7	19	5.00	25	15.377	0.980	-0.089	4.304	3.154	2.088	2064.402	0.394
2454	19	13	0.15	62	15.016	1.030	4.377	0.688	0.810	18.035	768.208	0.659
2464	11	12	0.95	20	9.844	2.325	-0.474	1.631	1.380	14.217	95.868	3.741
2478	13	11	0.95	30	13.636	0.673	-0.444	3.171	3.811	6.333	12004.117	0.770
2489	12	21	0.50	10	16.650	0.576	0.431	3.535	2.303	11.210	5663.716	0.843
2549	13	15	0.50	30	12.214	1.357	-1.055	1.409	1.157	14.758	520.220	2.891
2551	48	30	1.50	30	26.143	0.539	-0.286	3.207	4.141	6.669	93276.660	0.512
2667	55	67	30.00	50	16.977	1.695	0.758	3.919	3.803	1.184	3.650	5.430
2749	22	22	3.00	42	20.708	0.701	0.906	4.308	4.022	4.367	42318.566	0.340
2762	12	21	1.00	35	11.659	0.779	1.068	3.103	2.237	13.005	45.399	1.244
2860	9	6	1.00	30	12.735	0.816	0.133	3.096	3.728	6.933	22731.842	0.034
3064	13	5	0.20	25	14.106	0.630	-0.788	1.160	2.780	13.821	8157921.303	0.263
3080	25	27	0.09	25	23.442	0.531	-1.217	0.623	0.643	19.327	149115.198	0.187
3288	20	10	1.00	0	10.864	3.729	-1.169	0.822	1.750	15.448	3.518	0.616
3338	31	29	1.50	30	24.827	0.541	0.203	3.508	3.345	7.687	9639.132	0.244

3353	26	17	10.00	30	16.307	1.232	-0.249	3.749	4.305	2.066	23439.960	1.410
3366	16	11	0.50	50	15.653	0.693	-1.936	2.055	3.018	13.340	4647.231	0.499
3507	9	13	0.50	33	14.030	1.055	0.529	2.440	1.891	14.095	159.184	1.101
3509	8	13	0.20	30	12.714	0.741	-0.024	2.491	1.513	13.807	1802.071	0.884
3526	15	13	5.00	20	15.809	0.793	-0.053	4.127	4.159	1.280	6874.479	0.242
3641	7	41	1.50	43	20.406	0.579	0.615	4.292	1.875	3.434	2372.810	0.319
3774	29	26	0.50	50	23.554	0.537	-0.997	2.580	2.898	13.300	38.059	0.952
3800	21	28	3.00	15	20.046	0.820	3.384	4.322	3.599	6.632	1094.443	0.625
3926	25	20	0.30	25	22.016	0.547	-0.047	1.237	1.529	13.393	4272.601	0.332
3928	15	21	0.68	70	19.525	0.584	-0.421	3.604	3.051	11.838	81382.464	0.139
3929	35	31	8.00	50	5.865	3.653	0.062	3.165	3.520	10.320	4021.718	5.862
3960	47	50	8.00	40	28.973	0.883	-0.992	4.624	4.670	3.583	589.028	0.514
4104	8	6	20.00	20	10.864	2.116	0.036	3.729	3.994	0.742	36704.448	0.141
4366	31	31	8.00	2	23.414	0.913	-0.434	4.367	4.554	2.979	3592.689	0.487
4407	44	51	4.00	19	14.573	1.670	0.070	3.591	3.238	12.165	103.661	2.693
4453	15	18	2.00	30	17.165	0.617	1.204	4.110	3.071	2.607	44736.399	0.892
4477	7	11	2.00	50	11.870	1.104	-3.305	3.867	3.222	5.142	7249.061	0.325
4488	11	8	2.00	1	8.033	5.347	1.108	2.160	2.843	12.739	1301.149	1.559
4493	19	14	10.00	40	16.683	1.161	-0.191	3.883	4.306	1.567	7848.309	0.355
4657	66	62	4.00	30	34.581	0.559	0.330	4.657	4.691	3.410	1143492.186	0.205
4699	13	22	5.00	30	18.734	0.765	0.112	4.242	3.833	1.207	154786.619	0.211
4986	16	5	10.00	50	13.378	1.203	-0.029	3.287	4.306	1.214	66271.436	0.393
5056	18	35	5.00	33	21.957	0.720	-0.780	4.484	4.094	2.763	564328.562	0.324
5144	11	9	20.00	20	7.244	3.118	10.283	3.617	3.737	1.411	3589.357	2.695
5174	42	47	3.00	50	23.823	0.744	-1.240	4.346	4.162	8.369	233854.652	1.622
5229	28	31	15.00	10	12.332	3.417	0.320	3.961	3.564	6.028	12.149	2.880
5261	45	43	10.00	30	26.183	0.965	-1.578	4.361	4.651	2.525	81980.068	0.965
5357	36	22	5.00	30	22.768	0.774	1.093	4.028	4.506	3.022	12965.328	0.354
5390	22	17	4.00	30	17.724	0.819	0.513	3.650	4.317	4.262	1225.164	1.086
5497	44	24	4.00	30	25.292	0.652	0.642	4.231	4.571	3.005	11114241.101	0.233
6322	23	26	10.00	66	9.106	2.653	-0.405	3.753	3.536	5.852	59.531	3.981
6927	19	16	4.00	1	18.477	0.918	-0.601	3.925	4.247	4.178	8619.418	0.272
7115	19	39	4.00	30	15.827	0.848	0.444	4.314	3.226	3.401	314182.624	1.469
7162	51	14	1.00	50	20.424	0.808	-1.269	1.209	4.010	8.932	42025.044	1.093
7634	52	44	2.00	2	23.259	0.568	0.761	3.767	4.307	8.242	7551.190	1.509
7878	5	7	4.00	29	9.554	1.484	-0.612	3.815	3.285	1.450	360.702	0.875
8076	34	29	6.00	10	22.880	0.806	-1.387	4.321	4.419	2.903	10417.833	1.134
8078	29	32	4.00	36	8.066	1.796	1.664	3.587	3.088	10.469	112.042	5.352
8219	16	23	0.40	22	15.559	0.604	-0.087	2.601	1.940	14.337	726441.721	1.542
8396	53	59	0.50	1	13.606	1.385	7.016	0.880	0.767	16.517	1425.343	2.451
9120	52	44	10.00	20	28.586	0.876	-0.006	4.690	4.746	1.789	47120.478	0.219
9949	21	35	5.00	20	21.022	0.923	1.016	4.539	3.565	3.200	14751.471	0.563
10200	22	25	4.00	9	15.905	1.513	-7.127	3.619	3.583	6.466	34488.640	1.034
10684	16	58	4.00	57	18.929	0.659	-7.058	4.450	3.642	4.785	35.867	1.342
13333	18	19	2.00	25	19.948	0.638	0.168	3.975	3.913	6.246	77062.947	0.079
14242	68	37	3.00	20	31.135	0.558	0.383	4.266	4.663	5.096	2974.188	0.372
14420	56	20	20.00	22	13.393	1.571	-3.492	3.144	4.609	1.464	5.221	5.899
18045	40	50	4.00	30	26.932	0.654	-3.178	4.269	4.344	3.532	101.449	0.809
19018	63	67	8.00	30	29.425	0.835	1.891	4.634	4.360	2.918	18132.000	0.863
19991	22	50	4.00	20	24.460	0.658	0.062	4.599	3.990	3.472	934.865	1.058
20270	20	16	4.00	30	18.848	0.815	-0.255	3.707	4.250	3.938	285.531	0.531
21669	38	37	1.00	21	17.621	0.831	-1.649	1.917	2.520	14.959	2068.038	2.604
22962	23	18	4.00	24	17.168	0.879	-1.295	3.760	4.158	4.583	7451.091	1.143
23249	21	40	1.00	16	24.123	0.532	0.538	2.978	2.075	7.226	1758578.906	0.101
23652	24	24	1.00	20	20.607	0.643	-0.193	2.862	3.456	14.124	76202.298	1.056
37077	11	46	4.00	24	19.329	0.886	0.584	4.311	2.254	3.507	8747.907	1.563

38128	27	52	8.00	42	25.657	0.808	-0.562	4.687	4.444	1.655	25574.217	0.263
38760	10	17	5.00	33	12.412	1.155	0.026	4.192	3.255	2.936	5717587.740	2.551

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